# DESIGN AND EVALUATION OF GBBRPM: A GRAPH-BASED BOUNDED RISK PROPAGATION MODEL FOR COMPARATIVE PRIORITIZATION ON DIRECTED ACYCLIC GRAPHS ACROSS NETWORK DOMAINS

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## ACRONYMS AND ABBREVIATIONS

| | |
|---|---|
| API | Application programming interface |
| CSV | Comma-separated values |
| DAG | Directed acyclic graph |
| FKG | Fortuin–Kasteleyn–Ginibre |
| GBBRPM | Graph-Based Bounded Risk Propagation Model |
| IC | Independent Cascade |
| JSON | JavaScript Object Notation |
| LDAG | Local directed acyclic graph |
| MIA | Maximum influence arborescence |
| OSV | Open Source Vulnerabilities |
| RIPA | Risk impact pathway analysis |
| SIR | Susceptible-infected-recovered |
| SIS | Susceptible-infected-susceptible |
| SWMM | Storm Water Management Model |

# CHAPTER I
# INTRODUCTION

## 1.1. Background of the Study

Network propagation models represent connected entities and the relations through which one entity can influence another. Across applications, the propagated quantity may be activation, compromise, disturbance, or risk, and its meaning determines the appropriate update rule (Kempe et al., 2003; Muñoz-González et al., 2017; Faraji et al., 2026). A local observation and a receiving node's accumulated state are consequently different analytical quantities. When a model includes inherited influence, nodes with similar local inputs can acquire different receiving states because their predecessors and relation conditions differ. Comparative assessment therefore requires an explicit account of both the local state and the permitted network influence.

Several modeling choices shape this account. Graph topology determines which influences can reach a node; edge modifiers determine how strongly upstream state contributes; recursion determines whether inherited state continues along later relations; and aggregation determines how local and incoming terms combine. These choices can change magnitudes and rankings. Structural position, accumulated receiving state, influence exerted by a source, and benefit from intervention are distinct targets. An evaluation must identify its target before treating a high score as useful priority.

The mathematical components relevant to bounded propagation already have established precedents. The probabilistic-sum t-conorm combines bounded inputs through complement product (Klement et al., 2004). Leaky noisy-OR uses related background-plus-parent algebra in a probabilistic setting (Fallet-Fidry et al., 2012), while RIPA propagates current risk through relation and receiver modifiers with additive aggregation (Faraji et al., 2026). MIA restricts cascade calculations to local influence trees, and association-based cascade bounds provide another precedent for analyzing approximation (Chen et al., 2010a; Abbe et al., 2017). The research value of selecting these mechanisms will therefore depend on a justified formulation and comparative evidence. Boundedness alone establishes neither correct receiving-state rankings nor practical usefulness.

Shared ancestry provides a specific condition for investigating these choices. Incoming contributions may originate from different local sources or from one origin whose influence splits and reconverges. Separate aggregation of those terms can produce different results depending on the assumed process and dependence structure. Restricted-tree methods and cascade bounds explicitly address related approximation questions (Chen et al., 2010a; Abbe et al., 2017). The present study will examine how source uncertainty, attenuation, and graph structure affect the selected recursion's reference agreement and priority ordering, including regimes where simpler rules or dependence treatments perform equally well or better.

The study will specify and evaluate the **Graph-Based Bounded Risk Propagation Model (GBBRPM)**, the name assigned here to a particular architecture within graph-based risk propagation. Its core will combine local disturbance with modified current upstream states using the established complement-product operator on **directed acyclic graphs (DAGs)**. In domain applications, its output will be a dimensionless receiving-state index in $[0, 1]$. The core equations will be shared, while the meanings and construction of local disturbance, susceptibility, and transmission will remain specific to each domain. A **GBBRPM evaluation engine** will implement the versioned equations and expose their input and contribution traces through a research prototype.

The primary controlled evaluation will use a separately defined stochastic activation process with independent initial and conditional edge events. This construction will supply exact or uncertainty-qualified reference activation probabilities for matched comparisons. Under those assumptions, the GBBRPM recursion has a conditional nodewise upper-bound interpretation; Chapter 3 will state its bracket, proof, and relation to established bounding arguments. Experiments will evaluate bound tightness, fixed-budget receiving- node selection, and computational cost. This benchmark will characterize the mechanism under its specified process. Domain-index inputs will retain their own meanings, and practical usefulness will require independent outcomes appropriate to the selected domain task.

Urban drainage supplied the original practical motivation. Camera-based blockage assessment provides local condition information, while graph-based critical-pipe and weakest-link analyses evaluate connected-system behavior (Vandaele et al., 2024; Dastgir et al., 2023; Meijer et al., 2023). These approaches illustrate why local condition and network-context assessment deserve separate attention. The proposal will retain drainage alongside software-dependency and electrical- network cases as supporting instantiations of the common core. Their purpose will be to examine defensible mappings, DAG eligibility, and evidence limits. Three instantiations will not establish universal applicability, and their scores will not be pooled as a common physical risk scale. Preliminary worked calculations and exploratory findings will be retained in the appendices; the proposed evaluation protocols will be specified in Chapter 3.

## 1.2. Statement of the Problem

The central problem is to determine when GBBRPM's complement-product recursion gives useful receiving-state rankings under a declared reference process, how shared ancestry limits that agreement, and what computational tradeoffs accompany alternative mechanisms. Existing aggregation, restricted-tree approximation, and cascade-bounding methods establish relevant precedents (Klement et al., 2004; Chen et al., 2010a; Abbe et al., 2017). The proposed study will therefore assess the specified architecture against matched comparators rather than infer its contribution from a new name, a bounded formula, or different scores alone.

The controlled comparison will hold graph and primitive inputs fixed across methods while varying predeclared source, attenuation, and topology

conditions. Maximum and clipped-additive aggregation, local-only scoring, a recursion ablation, a restricted-tree comparator, and the ancestry-aware diagnostic will provide distinct comparison conditions. The analytic benchmark bracket will establish nodewise score bounds under independent primitive events; the experiments will determine their tightness and the resulting selection quality. A tighter score bound will not automatically imply a better ordering.

This evaluation will address a receiving-state task: accumulated local and inherited influence at candidate nodes. Its stochastic reference will define activation probabilities only within the constructed process. Domain indices will not be relabeled as event probabilities, and a high receiving-state score will not be interpreted as source influence or intervention benefit. Drainage, software-dependency, and electrical-network cases will assess coherent instantiation separately; any practical prioritization claim will require independent evidence relevant to that case's task.

The intended lightweight design will be assessed through matched runtime and memory measurements, required inputs, and preprocessing costs. A single-pass DAG core can have limited computational requirements while still omitting important dependence or physical mechanisms. The study will investigate that accuracy–cost tradeoff and report unfavorable or unresolved findings. It will address the following research questions:

1. How can local disturbance, current upstream state, relation-specific modifiers, and bounded aggregation be formulated into GBBRPM for comparative node-risk prioritization on DAGs?

2. How can the proposed formulation be implemented reproducibly in a GBBRPM evaluation engine and presented in a prototype that preserves input validation and computational traceability?

3. Under an explicitly specified stochastic cascade reference, how do GBBRPM, maximum aggregation, clipped additive aggregation, local-only scoring, a local-upstream recursion ablation, and a restricted-tree comparator differ in bound tightness and fixed-budget node selection?

4. How do shared ancestry, source uncertainty, and attenuation affect the receiving-state bracket's tightness and ranking quality; when does an ancestry-aware diagnostic improve or worsen them; and what mathematical and robustness properties characterize the frozen core?

5. What runtime, memory, and input requirements accompany GBBRPM, the matched comparators, and the reference solvers at stated precision?

6. Can the same core be instantiated coherently in drainage, software-dependency, and electrical-network cases, and what applicability limits follow from each case's data and evaluation?

## 1.3. Objectives of the Study

### 1.3.1. General Objective

The general objective is to specify, implement, and evaluate GBBRPM for bounded receiving-state prioritization on directed acyclic graphs, determine its reference accuracy and computational tradeoffs under controlled aggregation and dependence conditions, and establish the evidence limits of its drainage, software-dependency, and electrical-network instantiations.

### 1.3.2. Specific Objectives

Specifically, the study aims to:

1. formulate GBBRPM by specifying its DAG representation, bounded state semantics, local disturbance, relation-specific modifiers, recursive update rule, aggregation mechanism, assumptions, and expected mathematical properties;

2. implement a reproducible deterministic evaluation engine and an interactive research prototype that will validate inputs, execute the same versioned core, and expose traceable node states, edge contributions, and rankings;

3. evaluate receiving-state error and fixed-budget node selection against exact or uncertainty-qualified stochastic references using matched aggregation comparators, a recursion ablation, and a restricted-tree comparator, while separately characterizing heterogeneous-input effects and saturation;

4. characterize shared-origin dependence under controlled source and attenuation changes, evaluate an ancestry-aware diagnostic against the same reference, derive the benchmark bracket with its assumptions, and establish the frozen core's analytical properties, boundary behavior, sensitivity, and perturbation robustness;

5. characterize the evaluation engine's computational and input requirements through complexity analysis, runtime and memory measurements, and a documented comparison of required information and modeled mechanisms; and

6. assess whether the same core formulation can be instantiated coherently in the three selected domains through documented DAG construction, parameter mappings, within-domain experiments, and explicit evidence and applicability limits.

Literature synthesis will inform the design requirements and interpretation of these objectives. The principal research outputs will be the specified architecture, comparative evaluation evidence, and an account of accuracy, cost, and applicability conditions. The evaluation engine and prototype will support reproducibility and inspection of those outputs.

## 1.4. Significance of the Study

### 1.4.1. Theoretical Significance

The study will specify an architecture for combining local disturbance with recursively propagated network context on DAGs. The proposed methodological significance will lie in a reference-based account of when the adopted recursion gives accurate receiving-state estimates under stated stochastic assumptions, how shared origins affect that agreement, and what computational tradeoffs accompany alternative rules. Basic operator properties and the architecture's name will not stand as the original result. The declared index semantics will remain separate from benchmark probabilities. Under the synthetic independent-primitive process, the same GBBRPM recursion will be a nodewise upper bound; that conditional mathematical interpretation will not calibrate domain inputs. Existing association-based cascade bounds will be compared before any novelty claim (Abbe et al., 2017). The study will distinguish properties established by the formulation from claims that require computational or domain evidence, including the difference between bounded aggregation and correction for shared-origin dependence.

Existing propagation and risk literature supplies relevant mechanisms (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Muñoz-González et al., 2017; Petrone and Latora, 2018). The proposed contribution will be assessed through the specification and evaluation of the selected comparative-state task. The adopted t-conorm and its basic properties will be credited as established mathematics (Klement et al., 2004); their use or rederivation will not be claimed as a new aggregation theory. A common interface for domain-specific inputs will allow architectural reuse to be examined without assuming common physical semantics or universal applicability. Susceptibility and transmission will retain distinct conceptual roles, while their separate empirical identification will not be inferred from outputs that depend on their product.

### 1.4.2. Practical Significance

The evaluation engine and prototype will provide an inspectable workflow for examining local inputs, inherited contributions, and resulting receiving-state rankings. The comparison will characterize how the simple scalar recursion and alternative mechanisms differ in benchmark agreement and selected priorities. Runtime, memory, and input requirements will make the cost of those choices explicit.

The drainage, software-dependency, and electrical-network cases will document how the common architecture can be instantiated and where the available information restricts its interpretation. Potential screening or inspection uses will depend on defensible parameter construction and independent outcome evidence within each domain. Lower cost, plausible ordering, and coherent mappings will not alone establish maintenance benefit, exploitation prediction, or fault prediction. Operational decision claims will be limited to the tasks and outcomes actually evaluated.

## 1.5. Scope and Limitations

### 1.5.1. Scope of the Study

The study will cover the formulation and implementation of GBBRPM for static comparative node-risk assessment on DAGs. It will specify bounded local disturbance, susceptibility, optional transmission, edge contributions, node-state aggregation, topological evaluation, and tie-aware ranking. Controlled networks will support investigation of mathematical behavior, mechanism effects, robustness, shared-origin reconvergence, and computational cost. A separately specified stochastic cascade benchmark will provide exact small-graph references and Monte Carlo estimates with explicit sampling uncertainty. Matched shared- and independent- origin scenarios will be the main controlled dependence experiment; larger DAGs will support receiving-state and fixed-budget ranking comparisons. These planned experiments will be additional to the historical 136 runs. The bound and its proof will be conditional on the benchmark assumptions; probabilistic score bounds will not be asserted for domain-index outputs.

Three domain instantiations will be examined separately using documented topology and parameter mappings: drainage, software dependency, and electrical networks. The same core equations will be retained. The common object of investigation will be coherent instantiation of the architectural roles, rather than equality of physical outcomes or comparability of numerical scores across domains. Operational drainage or event-aligned electrical evaluation will proceed only if suitable data become available.

The research prototype will support validated synthetic and user-imported DAG inputs, controlled parameter editing, execution of the versioned evaluation engine, and traceable graph and ranking outputs. Functional verification will assess numerical consistency, input handling, and presentation of the computation.

### 1.5.2. Limitations of the Study

The core will be restricted to DAGs and one-pass evaluation. It will not represent cycles, bidirectional feedback, physical-time evolution, or full hydraulic or electrical dynamics. A topological order is a computational dependency order, not a physical propagation timeline. The index will not be a calibrated probability, and the local aggregator will not correct shared-origin dependence. Susceptibility and transmission will not be identifiable separately from output states alone. The stochastic benchmark will quantify agreement only with its declared independent primitive process. It will not establish that advisory severity, hydraulic loading, or electrical conditions are calibrated event probabilities. The ancestry-aware diagnostic may discard legitimate cumulative effects and will not be presumed superior.

Interpretation will depend on the provenance, coverage, and defensibility of domain inputs. Synthetic networks and simulated hydraulic outputs will not be treated as field observations. Advisory-based software mappings will not establish exploitation probabilities. Controlled disturbances applied to measured electrical loading will not establish event-response prediction. The currently available electrical baseline is not aligned with independent outcomes for the

documented switching windows. Inadequate field records or event-aligned outcomes may restrict evaluation to structural instantiation and controlled behavior.

Higher-fidelity formulations may be required for feedback, shared dependencies, and physical state interactions (Buldyrev et al., 2010; Shuang et al., 2017; Petrone and Latora, 2018; Wu et al., 2026; Dastgir et al., 2024; Zhang et al., 2024). The study will report these omissions as tradeoffs rather than presume that reduced computational cost preserves predictive fidelity. Prototype checks will establish consistency in the tested configuration, not production security, multi-user readiness, usability, or operational utility.

## 1.6. Conceptual Framework

The conceptual framework of the study begins with local disturbance and directed acyclic network context. The current risk state at an upstream node is transformed into a relation-specific contribution according to susceptibility and, where justified, optional transmission. Multiple incoming contributions are then combined with any local disturbance at the receiving node using a bounded aggregation rule, producing an updated comparative risk state. That updated state can in turn influence later downstream relations, allowing the modeled risk state to evolve recursively through the network. The resulting node states provide the basis for comparative prioritization.

Figure 1.1 distinguishes the domain-supplied inputs, the generic bounded propagation core, and the comparative decision-support output.

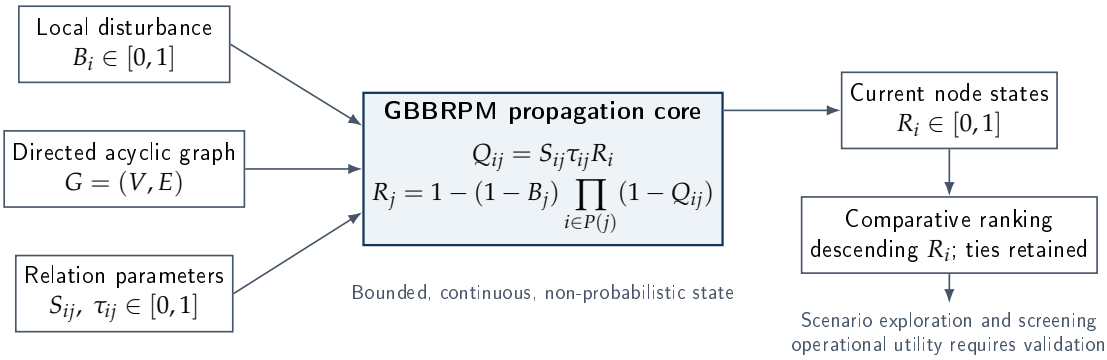



**Figure 1.1.** Conceptual framework of GBBRPM, distinguishing supplied domain inputs, the generic propagation core, and the comparative decision-support output.

For a directed relation $i \to j$, the generic edge contribution is:

$$Q_{ij} = S_{ij}\tau_{ij}R_i \qquad \textit{(Equation 1-1)}$$

where $R_i$ is the current upstream normalized risk state, $S_{ij}$ is relation-specific susceptibility, and $\tau_{ij}$ is optional transmission or influence.

The receiving-node state is:

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}) \qquad \text{(Equation 1-2)}$$

where $B_j$ is the local disturbance and $P(j)$ is the set of predecessors of node $j$. This is the established probabilistic-sum t-conorm extended to the local and incoming inputs (Klement et al., 2004). Its adoption supplies bounded saturating aggregation, not a newly invented operator. Despite the related noisy-OR algebra, this index is not interpreted as a probabilistic union because its inputs are not calibrated probabilities.

The resulting $R_j$ is the node's current modeled receiving state, including local and inherited disturbance. It is not an estimate of the benefit from mitigating that node. It can serve as the upstream state for later relations, so subsequent calculations retain the accumulated influence. Node ranking will be derived from the final receiving states rather than treated as an additional propagation mechanism.

A separately specified stochastic reference will support controlled evaluation of the recursion; domain usefulness will require independent outcomes rather than agreement with the index itself.

### 1.7. Definition of Terms

| Term | Definition |
|---|---|
| Comparative risk prioritization. | Ranking components according to their modeled risk states relative to one another under the same network and parameterization. |
| GBBRPM. | Graph-Based Bounded Risk Propagation Model, the name assigned in this study to its proposed bounded, non-probabilistic comparative-risk model on DAGs; not an established model-family name. |
| GBBRPM evaluation engine. | Software that will validate inputs and apply the versioned GBBRPM equations in topological order; the implementation of the model. |
| Directed acyclic graph (DAG). | A directed graph containing no directed cycle. This is the supported graph class for the proposed one-pass engine. |
| Lightweight model. | A fixed-form, single-pass DAG model without physical-time integration or training in its core. Runtime, memory, preprocessing, and input requirements will characterize this design; reduced cost does not establish physical fidelity. |
| Stochastic reference. | A separately defined activation process with independent initial and conditional edge events, used to evaluate the recursion under declared probabilistic assumptions; not a calibration of the domain risk index. |

| Term | Definition |
|---|---|
| Shared ancestry. | Overlap between the local-origin lineages reaching different incoming relations. Structural overlap alone does not specify the magnitude of event dependence. |
| Receiving-state ranking. | Ordering nodes by accumulated local and inherited modeled disturbance; distinct from ranking the benefit of intervention at those nodes. |
| Preliminary evidence. | Completed exploratory calculations or tests used to assess feasibility and limitations before protocol freezing, distinguished from independent confirmatory or outcome validation. |
| Normalized risk state. | A dimensionless continuous value in $[0, 1]$ representing the current modeled level of network-context risk at a node. It is not a calibrated probability of an event. |
| Local disturbance. | A bounded input representing risk or disturbance introduced directly at a node according to the selected domain. |
| Susceptibility. | A bounded relation-specific factor representing how strongly a relation or receiving context responds under the evaluated condition. |
| Transmission or influence. | An optional bounded factor describing how strongly influence transfers along an existing directed relation. |
| Edge contribution. | The relation-specific contribution generated from current upstream risk, susceptibility, and optional transmission. |
| Topology or reachability. | The directed network structure that determines which nodes and relations can participate in propagation. |
| Bounded aggregation. | Aggregation through the established probabilistic-sum t-conorm, combining local disturbance and incoming contributions while preserving the $[0, 1]$ index range. |
| Domain-specific instantiation. | Defining the shared node, edge, disturbance, susceptibility, and transmission roles with defensible meanings and inputs for a particular domain. |
| Generic formulation. | A core formulation not hard-coded to one domain. Reusability of its architecture does not establish universal empirical applicability. |

# CHAPTER II
# REVIEW OF RELATED LITERATURE

## 2.1. Review Basis and Organization

This chapter synthesizes two complementary evidence streams. The first concerns urban drainage monitoring, blockage, graph-based vulnerability assessment, critical-component analysis, and hydraulic or network-level prioritization. This literature establishes the practical origin of the research problem. The second concerns foundational and modern propagation models across influence diffusion, epidemics, cascading failure, cybersecurity, finance, transportation, and infrastructure. This literature establishes the mechanisms and state semantics that can inform a general graph-risk architecture. Within this study, drainage literature provides motivating evidence and one application context. The general mechanism- selection question introduced in Chapter 1 remains the study's scientific focus; the review's domain-first order does not designate drainage as the primary outcome-validation case.

The review follows a two-stage synthesis. It first establishes the domain-specific problem through drainage monitoring and graph-based prioritization. It then situates that problem within the broader literature on propagation-state semantics, topology, transmission, susceptibility, recursion, aggregation, dependence, model complexity, and ranking. The final sections compare reusable architectures and domain-specific parameterization before synthesizing the research problems and the design and evaluation question addressed by the study.

## 2.2. Local Assessment and Structural Network Prioritization

The following studies establish why local condition and network importance are different analytical targets. They move from drainage connectivity to asset-level monitoring and then to structural and condition-aware ranking.

### 2.2.1. Urban Drainage Networks as Interconnected Systems

Urban drainage networks consist of connected conduits, junctions, inlets, screens, and other components whose behavior is influenced by flow direction, network structure, hydraulic condition, and local impairment. Blockage, clogging, siltation, structural defects, or reduced effective capacity can influence system behavior beyond the initially affected location. Numerical and network-based drainage studies show that the location and severity of impairment can alter hydraulic response, vulnerability, and the importance of connected components (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

This interconnected structure creates a practical prioritization problem. Inspection and maintenance resources are finite, so not every component can receive equal attention at the same time. Existing drainage studies explicitly

frame critical-component identification as useful for targeted management, maintenance, monitoring, resilience, or dredging decisions (Simone et al., 2022), (Simone, 2023), (Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). The practical decision problem is therefore not only to identify which components are locally impaired but also to determine which components deserve greater attention when the surrounding network is considered.

This drainage-side evidence exposes a distinction between observation and prioritization. A local blockage or defect can be observed directly, but a maintenance decision often requires a broader question: whether that component remains important after the connected network, downstream pathways, relation conditions, and converging influences are considered. This distinction between local condition and network-context priority recurs in studies that combine inspection, topology, capacity, and hydraulic relevance (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

### 2.2.2. Local Monitoring and Component-Level Condition Assessment

Computer-vision, sensor, and inspection studies have substantially improved the ability to identify local drainage condition. Iqbal et al. (2021), for example, formulate culvert blockage as an image-classification problem and evaluate several convolutional neural networks for classifying blockage from imagery. Vandaele et al. (2024) similarly develop camera-based methods for automatically detecting blockage at trash screens for maintenance and flood-management use. These approaches demonstrate that local blockage information can be generated automatically and repeatedly at monitored assets.

The outputs of these methods are valuable because they provide repeatable local condition or disturbance signals. However, a local score by itself does not determine network-context priority: it describes the monitored asset rather than the recursively accumulated state of the wider connected system (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022).

This distinction matters when two components have similar observed local conditions but occupy different positions in the network. One may lie on a relation with high operating susceptibility or feed several important downstream paths, while another may have little propagated consequence under the same network configuration. Local assessment is therefore necessary for condition awareness, but it does not by itself answer which nodes warrant greater priority after network position and operating conditions are considered (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2025).

The limitation should not be interpreted as a weakness of computer vision or sensing. Those methods solve the detection problem they were designed to solve. The unresolved issue is the interface between local condition information and network-level prioritization. This motivates the need for a propagation layer that can accept a local disturbance as input while separately representing network structure and relation-specific response (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023).

### 2.2.3. Graph-Based Drainage Analysis and Structural Prioritization

Drainage research has increasingly used graph theory and complex-network analysis to identify critical or vulnerable components. Simone et al. (2022) examine standard and tailored centrality metrics for vulnerability assessment, monitoring design, and contaminant or pathogenic spread. Their analysis shows that standard degree, harmonic, betweenness, and edge-betweenness measures do not always reproduce hydraulically meaningful importance, which motivates tailored metrics that incorporate intrinsic relevance and flow direction.

Simone (2023) extends this idea using relevance-based centrality for critical-pipe identification. The method ranks nodes and pipes by combining network connectivity with intrinsic relevance rather than using plain topological centrality alone. This is significant for the present study because it demonstrates that even within a graph-only framework, useful prioritization can require information beyond raw connectivity.

Dastgir et al. (2023) propose a graph method for critical-pipe analysis based on runoff edge betweenness and capacity-related information. Their method is compared with hydrodynamic simulation and is intended to retain useful criticality information with lower computational and data requirements. The study also reports that graph-method accuracy decreases as network complexity, loop degree, and hydraulic stress increase, illustrating the tradeoff between lightweight structural approximation and richer physical behavior.

Dastgir et al. (2024) subsequently examine multiple-pipe failures in stormwater networks using graph-theoretic representations. Their work is particularly relevant to the present study because it shows that the consequences and prioritization of interacting failures cannot always be inferred from isolated single-component analysis. It therefore supports the explicit evaluation of multi-source configurations and converging effects while also reinforcing the need to distinguish graph-based screening from full hydraulic consequence prediction.

Meijer et al. (2023) develop a graph-based weakest-link method for urban drainage systems using flow-path analysis instead of repeated hydrodynamic simulation. Their work focuses on storage and discharge-capacity relationships and demonstrates that graph-based screening can make long rainfall-series analysis practically feasible. This provides further evidence that drainage research already seeks computationally efficient prioritization and vulnerability tools rather than relying exclusively on full hydraulic simulation.

Zhang et al. (2025) analyze siltation risk using complex-network concepts, centrality, pipeline risk factors, and edge vulnerability. Their study identifies critical nodes and high-risk propagation pathways, explicitly linking siltation-chain structure with maintenance and dredging prioritization. This is especially relevant because it shows that drainage risk prioritization may involve both structural importance and domain-specific condition information rather than a single metric.

Taken together, these studies establish two important points. First, prioritization is already a legitimate drainage-network task; the proposed contribution of the present study is therefore not the act of ranking nodes or pipes. Second, the literature repeatedly augments topology with runoff, hydraulic relevance, storage/discharge capacity, edge vulnerability, siltation condition, or other op-

erational information. Structural position and operating response are therefore related but not equivalent (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

### 2.3. Propagation Models and State Semantics

Broader network science provides several foundational propagation paradigms. Independent Cascade and Linear Threshold models represent influence through activation semantics (Kempe et al., 2003). Epidemic models such as SIR and SIS propagate compartment or infection states (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021). Cascading-failure models propagate physical load, capacity exceedance, or component failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Attack graphs represent feasible attack paths and reachable compromise states, while Bayesian attack-graph approaches introduce probabilistic compromise semantics (Sheyner et al., 2002), (Muñoz-González et al., 2017). Cyber-risk, finance, transportation, and infrastructure models introduce additional states such as accumulated risk, exposure, payment loss, systemic stress, or dynamic load (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018).

The main lesson is that the meaning of the propagated state determines the valid update and aggregation rules. Binary activation answers whether a node changes state. Probability estimates event likelihood under a probabilistic model. Physical-load models track conserved or redistributed quantities. These states are appropriate for their intended questions, but they do not automatically answer the comparative question of which node carries greater comparative network-context risk under a common parameterization (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Kavallieratos et al., 2021).

The comparison indicates that any proposed propagation model must declare whether its state represents activation, probability, physical quantity, or a non-probabilistic index. Without that declaration, similarly bounded values may be interpreted incorrectly and rankings from different model families may appear comparable when they are not (Kempe et al., 2003; Kermack and McKendrick, 1927; Muñoz-González et al., 2017; Petrone and Latora, 2018).

Table 2.1 compares the model families by their structure, propagated state, update basis, analytical purpose, and limitations to assess suitability for a lightweight comparative-risk task. The comparison shows why no single family can be adopted merely because it uses a graph; the semantics of its state and the assumptions of its update rule must match the research question.

**Table 2.1.** Comparison of propagation-model families relevant to comparative network-risk prioritization.

| Model family | Typical state | Propagation or update basis | Principal analytical use | Limitation for the present problem |
|---|---|---|---|---|
| Influence diffusion (Kempe et al., 2003) | Binary activation or threshold state | Edge influence probabilities or accumulated neighbor influence | Spread maximization and activation reach | Activation semantics do not directly represent a continuous comparative risk index. |
| Epidemic propagation (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021) | Susceptible, infected, and recovered compartments or infection probabilities | Contact structure, infection rate, and recovery rate | Disease spread and intervention analysis | Epidemiological compartments and rates require domain-specific assumptions not implied by network priority. |
| Cascading failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017) | Load, capacity, operational state, or failure state | Load redistribution, tolerance, and interdependence | Robustness and cascade consequence | Physical load conservation or failure thresholds are not generic comparative-risk semantics. |
| Attack graph and Bayesian attack graph (Sheyner et al., 2002; Muñoz-González et al., 2017) | Reachable compromise state or compromise probability | Feasible attack steps, conditional probabilities, or logical relations | Security-path and compromise analysis | Probabilistic outputs require calibrated conditional assumptions and dependence treatment. |
| Financial and infrastructure contagion (Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026) | Exposure, loss, stress, or service degradation | Dependency, exposure, vulnerability, and feedback | Systemic-risk and resilience assessment | Feedback and common exposure can require cyclic or dependence-aware dynamics. |

**Table 2.1.** Comparison of propagation-model families relevant to comparative network-risk prioritization (continued).

| Model family | Typical state | Propagation or update basis | Principal analytical use | Limitation for the present problem |
|---|---|---|---|---|
| Graph-based drainage screening (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025) | Centrality, criticality, capacity, or vulnerability score | Topology combined with relevance, runoff, capacity, or siltation information | Monitoring, maintenance, and critical-component screening | Scores differ in meaning and may omit recursive current-state propagation. |
| Hydraulic simulation and learned surrogate (Reyes-Silva et al., 2020; Dastgir et al., 2024; Zhang et al., 2024) | Flow, depth, surcharge, flood volume, or learned hydraulic state | Governing physics or simulator-trained approximation | Physical consequence analysis and prediction | Greater data, calibration, or training requirements; outputs answer a different question from lightweight comparative risk. |

## 2.4. Mechanisms Relevant to Comparative Propagation

Having distinguished the states used by existing model families, the review examines the mechanisms that shape propagation: permitted paths, relation modifiers, recursion, aggregation, and dependence.

### 2.4.1. Topology, Reachability, and Transmission

Across propagation models, topology primarily determines which interactions are possible. An edge $i \rightarrow j$ indicates that influence may move from node i to node j, but the existence of the edge does not by itself define the strength of influence. Influence-diffusion models, attack graphs, cyber-risk models, and weighted infrastructure models commonly distinguish the existence of a pathway from the weight, probability, exposure, or transfer mechanism associated with that pathway (Kempe et al., 2003; Sheyner et al., 2002; Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

This distinction is particularly important for interpreting centrality. Centrality uses global graph structure and can identify structurally important nodes or edges, but it does not directly simulate the propagation of a current disturbance under current relation-specific conditions. Simone et al. (2022) provide a drainage-specific example: standard centrality metrics can rank structurally cen-

tral elements differently from hydraulically relevant elements, which motivates the incorporation of relevance and direction.

Accordingly, the literature supports treating topology as reachability while retaining transmission or influence as a conceptually separate relation-level mechanism. Where heterogeneous influence cannot be empirically justified, a neutral coefficient is more defensible than an unsupported set of edge weights (Kempe et al., 2003; Simone et al., 2022; Dastgir et al., 2023).

### 2.4.2. Susceptibility, Capacity, Vulnerability, and Receiver Response

A second distinction concerns transmission versus susceptibility. Transmission describes how strongly an influence transfers along a relation, whereas susceptibility describes how strongly a relation or receiving context responds under the current condition. Similar distinctions appear in models using capacity, vulnerability, tolerance, resilience, exposure, or fragility (Buldyrev et al., 2010; He et al., 2019; Kavallieratos et al., 2021; Faraji et al., 2026).

Cascading-failure models demonstrate that the same incoming load can produce different outcomes depending on component capacity or tolerance (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Cyber and infrastructure models likewise separate dependency or exposure strength from vulnerability or resilience (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023). Drainage studies that incorporate hydraulic relevance, runoff, capacity, storage/discharge relationships, siltation condition, or edge vulnerability similarly show that structural connectivity alone is not sufficient to characterize operating response (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

These patterns support keeping relation-specific susceptibility distinct from transmission rather than collapsing both concepts into one unexplained coefficient. In a drainage application, susceptibility may be linked to load relative to effective capacity, whereas heterogeneous transmission should remain neutral unless empirical evidence justifies its calibration (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

### 2.4.3. Recursive Propagation and Current-State Updating

Many propagation systems are recursive: later effects depend on the state produced by earlier propagation rather than repeatedly referencing only the original source. Epidemic models update infection state over successive interactions, cascading-failure models redistribute load after each failure, financial-contagion models update exposure or loss as conditions change, and hydraulic models update system state in response to preceding conditions (Kermack and McKendrick, 1927; Buldyrev et al., 2010; He et al., 2019; Faraji et al., 2026; Petrone and Latora, 2018).

This pattern supports current-state recursion: for a chain $A \rightarrow B \rightarrow C$, the contribution from B to C should be based on the current modeled state at B when the intended semantics permit inherited and local influence to accumulate. Conceptually, the node propagates what it currently carries rather than only the disturbance that originally entered the network (Kermack and McKendrick, 1927; Buldyrev et al., 2010; Faraji et al., 2026).

Current-state recursion is central to comparative network context. If the model repeatedly used only the original local disturbance, intermediate accumulation would be lost and downstream ranking would not reflect the evolving state of the network (Kempe et al., 2003; Buldyrev et al., 2010; Petrone and Latora, 2018).

### 2.4.4. Multi-Source Aggregation and Bounded State

When several upstream relations converge on the same node, their contributions require aggregation. Existing model families use different operators according to state semantics: physical loads may be summed or redistributed, threshold models accumulate influence until a threshold is crossed, and probabilities may be combined through Bayesian or noisy-OR structures when the required assumptions are satisfied (Kempe et al., 2003; Muñoz-González et al., 2017; Buldyrev et al., 2010).

For bounded inputs, established t-conorms provide alternatives to unrestricted addition. A t-conorm is commutative, associative, monotone, and has zero as a neutral element. The probabilistic sum is

$$\mathcal{S}_P(x, y) = x + y - xy = 1 - (1 - x)(1 - y). \qquad \textit{(Equation 2-1)}$$

The maximum and bounded-sum t-conorms are also established (Klement et al., 2004). Associative repetition of Equation (2-1) yields $1 - \prod_k (1 - x_k)$. Its standard name does not require every application to assign probabilistic semantics, nor does selecting it establish a new aggregation operator.

These alternatives represent different accumulation behavior. Maximum retains the strongest input; bounded sum reaches one when its sum reaches the cap; and probabilistic sum combines nonzero inputs with a gradual approach to saturation. Basic boundedness and monotonicity are therefore insufficient to justify one choice over the others (Klement et al., 2004). Selection requires a declared index interpretation and examination of the resulting response and ranking behavior.

### 2.4.5. Close Mathematical and Risk-Architecture Precedents

Leaky noisy-OR provides a close background-plus-parent algebraic precedent. In an efficacy-parameter representation, its conditional probability has the form

$$\Pr(Y = 1 \mid \mathbf{X} = \mathbf{x}) = 1 - (1 - \ell) \prod_{i:x_i=1} (1 - p_i), \qquad \textit{(Equation 2-2)}$$

where $\ell$ accounts for a leak cause and $p_i$ are causal-effect parameters under the model's assumptions. Fallet-Fidry et al. develop evidential extensions for uncertainty in these structures (Fallet-Fidry et al., 2012). A bounded local input can occupy an analogous algebraic position, but it need not mean an unmodeled probabilistic cause. A forward index recursion using scalar upstream states is not automatically equivalent to Bayesian inference over correlated parents.

RIPA is a close architectural precedent for propagating current total risk through multiplicative relation and context factors. Its published propagation

and node aggregation rules are

$$R_{\text{pro},ij} = R_{\text{total}}(i)(1 - RE(i))(1 - RE(j))LI(i,j)C(j),$$
$$R_{\text{total}}(j) = R_0(j) + \sum_i R_{\text{pro},ij}. \quad \text{(Equation 2-3)}$$

Here the factors represent resilience, linkage intensity, and receiver criticality (Faraji et al., 2026). Current-state updating and multiplicative modifiers therefore already have direct risk-model precedents. The additive versus bounded aggregation choice is a comparison dimension, not evidence that all other architectural features are new.

**Table 2.2.** Close mathematical and architectural precedents and the comparison required.

| Precedent | Established overlap | Distinction or evaluation need |
|---|---|---|
| Probabilistic-sum t-conorm (Klement et al., 2004) | Exact repeated complement-product aggregation; established boundedness, associativity, and monotonicity | Identifies adopted mathematics. A graph-index application must justify its semantics and aggregation choice; it cannot claim a new operator. |
| Leaky noisy-OR and evidential extensions (Fallet-Fidry et al., 2012) | Background term plus parent influences in a complement-product structure | Conditional/evidential uncertainty differs from a supplied scalar index. Similar algebra does not equate index recursion with Bayesian inference. |
| RIPA (Faraji et al., 2026) | Current total-risk propagation through multiplicative context and dependency factors, with initial plus incoming risk | Uses additive node aggregation. Comparing accumulation behavior does not establish superior ranking or operational accuracy. |

The comparison separates operator equivalence, architectural overlap, state interpretation, and study evidence. A different model name, DAG restriction, non-probabilistic label, prototype, or selected combination of cases will not by itself establish mathematical novelty. The review will need to consider equivalent formulations under other names; the selected comparisons are not an exhaustive database search.

### 2.4.6. Cascade Bounds and Restricted Influence Structures

The stochastic-reference direction introduces additional close prior art. Abbe et al. (2017) derive cascade bounds using FKG inequalities and nonbacktracking message passing. The expanded journal treatment (Abbe et al., 2020) defines the activation probability of each node before summing those probabilities into seed-set influence. Its upper construction combines length-indexed incoming messages by complement product (Section 3.1, Equations 6–8). Its lower construction processes a directed acyclic subnetwork in topological order with an incoming-message inequality (Section 4.1, Equations 21–23), including

an $O(|V| + |E|)$ implementation. Thus nodewise bounds, positive-association arguments, and efficient DAG lower bounds already have clear precedents; the aggregate influence target does not erase their relevance to receiving-node evaluation.

The present benchmark uses independently sampled local activations, a single topological complement-product pass for $R_j$, a recursive maximum lower bound, and an optional target-specific restricted tree. Abbe et al.'s upper messages are indexed by path length, while their lower recurrence and subnetwork restriction differ from these two proposed lower comparators. An auxiliary certain seed with independent edges of weights $B_j$ can encode the random local starts, so that input convention alone will not be presented as a novelty distinction. The published DAG lower recursion will be tested as a separate matched nodewise comparator after this transformation in Section 3.6.5. No ordering between its lower values and $M_j, T_j$ will be assumed. The intended result is a transparent DAG specialization and measured tightness, ranking, and computational tradeoffs under this declared benchmark; any stronger novelty claim will depend on the final comparison and evidence.

Chen et al. (2010a) construct maximum-influence in-arborescences by retaining consistent maximum-probability paths and applying an influence threshold. Their Algorithm 2 evaluates activation through complement-product recursion on a restricted tree, where multiple seed influences can be aggregated. The tree removes the dependency issue associated with alternative paths from the same origin. MIA therefore provides both a mathematical precedent and a relevant restriction-based comparator; it is not equivalent to retaining one largest contribution globally. Its complete algorithm selects seeds for influence maximization, whereas this study fixes local inputs and evaluates receiving-state queries.

Chen et al. (2010b) instead develop local DAG influence computation under the Linear Threshold model. Their DAG activation recurrence is additive under that model's incoming-weight constraints. LDAG is consequently related structural approximation literature, not an interchangeable Independent Cascade reference or the same ancestry-grouping algorithm.

These studies will be distinguished from the prospective diagnostic that groups overlapping local-origin label sets and keeps one group maximum. Overlap grouping does not construct target-specific arborescences and does not automatically supply exact restricted-process probabilities. The study will compare its assumptions and failure cases with the tree restriction before any extension is described as original. A targeted literature review of equivalent approximations will continue; neither the bracket proof nor the proposed evaluation protocol is a verified priority claim.

### 2.4.7. Dependence, Shared Paths, and Reconvergence

Bounded aggregation does not solve every multi-source problem. When contributions share a common origin and travel through multiple paths that later reconverge, incoming contributions may not be independent descriptions of distinct risk sources. Similar dependence problems appear in financial networks through common exposures, in infrastructure networks through shared dependencies, and in graph systems with overlapping paths (Buldyrev et al.,

2010; Faraji et al., 2026; Zhou et al., 2023).

The literature therefore supports treating local aggregation and dependence as separate mechanisms. Bounding incoming contributions does not by itself correct shared-origin or common-cause dependence; reconvergence should be evaluated as an explicit limitation rather than hidden inside the aggregation rule (Buldyrev et al., 2010; Faraji et al., 2026).

This distinction is important for interpretability. A bounded output does not automatically mean that shared-origin information has been counted correctly. Dependence-aware correction remains a separate extension problem (Muñoz-González et al., 2017; Faraji et al., 2026).

## 2.5. Existing Ranking and Prioritization Approaches

Existing drainage and network studies rank components using different criteria. Local monitoring may rank according to observed blockage or defect severity. Centrality approaches rank according to structural importance. Weakest-link and vulnerability methods incorporate combinations of topology and domain-specific information. Failure studies rank components or combinations according to modeled consequence. Siltation studies identify high-risk nodes and propagation pathways. Hydraulic approaches may prioritize locations according to depth, surcharge, flood volume, resilience, or performance loss (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

The evidence shows that ranking is not a single modeling operation with a universal meaning. A centrality rank, blockage-severity rank, weakest-link rank, hydraulic-impact rank, and propagated-risk rank answer different questions. The semantic basis of the score must therefore be explicit before rankings can be compared (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

**Table 2.3.** Comparison of ranking and prioritization approaches

| Approach | Primary ranking basis | What the ranking answers | Limitation relative to the present target |
|---|---|---|---|
| Local condition / detection (Iqbal et al., 2021; Vandaele et al., 2024; Patil et al., 2023) | Observed blockage, defect, debris, or local score | Which asset appears more locally impaired? | Does not inherently incorporate recursively propagated network context |
| Centrality / structural criticality (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020) | Graph position, paths, relevance, or structural interaction | Which node or edge is more structurally important? | Does not by itself represent propagation of a current disturbance under current relation conditions |

| Approach | Primary ranking basis | What the ranking answers | Limitation relative to the present target |
|---|---|---|---|
| Weakest-link / vulnerability methods (Dastgir et al., 2023; Meijer et al., 2023) | Structural information combined with capacity, storage, discharge, redundancy, or other domain variables | Which component most constrains or weakens system performance? | Produces specialized vulnerability or criticality semantics rather than a generic bounded recursive risk state |
| Siltation-chain risk (Zhang et al., 2025) | Pipeline risk, node importance, edge vulnerability, and propagation path | Which nodes and pathways are high priority for siltation management? | Domain-specific siltation semantics and parameterization |
| Hydraulic consequence analysis (Dastgir et al., 2024; Zhang et al., 2024) | Simulated depth, surcharge, flood volume, performance loss, or resilience | Which failures or locations cause greater physical consequence? | High-fidelity but strongly domain-specific and more data/computation intensive |

The literature comparison also separates score generation from the decision-support use of a score. A method may first compute local, structural, hydraulic, or propagated values and then order components for inspection or maintenance; the resulting ranking inherits the meaning and limitations of the underlying score (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

### 2.5.1. Drainage Prioritization Evidence Matrix

The following study-level comparison makes the progression from local condition assessment to structural, hybrid, risk-path, and learned hydraulic approaches explicit.

**Table 2.4.** Drainage prioritization evidence matrix

| Study | Primary task | Representative information used | Prioritization / output | Relevance to the present problem |
|---|---|---|---|---|
| Iqbal et al. (2021) | Culvert blockage classification | Image-derived visual features | Local blockage class | Provides local condition information but not propagated network context |

| Study | Primary task | Representative information used | Prioritization / output | Relevance to the present problem |
|---|---|---|---|---|
| Vandaele et al. (2024) | Trash-screen blockage monitoring | Camera imagery and learned blockage features | Asset-level blockage information for maintenance/flood management | Demonstrates operational value of automated local disturbance information |
| Simone et al. (2022) | Vulnerability and monitoring analysis | Topology, intrinsic relevance, flow direction | Tailored node/edge centrality ranking | Shows that raw topology alone may not represent functional importance |
| Simone (2023) | Critical-pipe identification | Connectivity and relevance-based centrality | Pipe criticality ranking | Supports network-aware prioritization but not current disturbance propagation |
| Dastgir et al. (2023) | Critical-pipe analysis | Runoff paths, edge betweenness, capacity-related information | Critical-pipe ranking compared with hydrodynamic results | Demonstrates lightweight graph screening and its accuracy/complexity tradeoff |
| Meijer et al. (2023) | Weakest-link analysis | Storage, discharge capacity, and graph flow paths | Subsystem weakness / flood-performance screening | Shows that prioritization may combine structure with operating-capacity information |
| Zhang et al. (2025) | Siltation-chain risk analysis | Pipeline risk factors, centrality, edge vulnerability, propagation paths | Critical nodes and high-risk pathways | Closest drainage precedent for propagated risk-path prioritization, but with siltation-specific semantics |
| Zhang et al. (2024) | Hydraulic-state surrogate prediction | Graph topology, recent hydraulic states, runoff, control/boundary information | Predicted hydraulic variables rather than a direct risk ranking | Defines the learned/high-fidelity boundary of the proposed lightweight risk-index model |

The matrix shows that existing studies already provide local condition signals, graph-based criticality, capacity-aware screening, propagation pathways, and richer hydraulic state modeling. These strands remain distributed across methods with different purposes and state semantics, which creates a mechanism-selection question rather than an absence of ranking methods (Iqbal et al., 2021; Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

## 2.6. Computational and Input Tradeoffs

The preceding ranking approaches answer different questions. Their computational and data requirements must therefore be considered alongside their state semantics rather than treated as interchangeable measures of performance.

### 2.6.1. Lightweight Graph Models, Learned Surrogates, and High-Fidelity Simulation

The drainage literature contains a spectrum of model complexity. Local scores and graph indicators provide relatively inexpensive screening; graph-hydrodynamic hybrids combine structural reasoning with physical analysis; hydraulic simulators explicitly represent domain dynamics; and graph-neural-network surrogates learn simulator-derived hydraulic behavior for faster inference (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Dastgir et al. (2023) explicitly compare a graph-based critical-pipe method with hydrodynamic modeling and report substantially lower data and computational requirements for the graph approach, while also documenting accuracy loss as complexity and stress increase. Simone et al. (Simone et al., 2022) likewise position tailored complex-network metrics as complementary preliminary tools when full hydraulic simulation is more complex, data demanding, or computationally expensive.

At the learned-surrogate end of the spectrum, Zhang et al. (2024) develop a graph-neural-network surrogate for urban-drainage hydraulic prediction. The model is trained against physics-based flow-routing outputs and predicts hydraulic variables for real-time use. Physics-guided mechanisms are included to improve prediction consistency. Such a surrogate seeks to emulate hydraulic state transitions, which is a different objective from transparent graph screening or comparative index construction (Zhang et al., 2024).

The literature therefore does not imply that lightweight models are universally superior. High-fidelity models provide richer physical detail, while learned surrogates can approximate that detail after training. Lightweight graph models instead trade physical completeness for lower data requirements, simpler computation, and direct interpretability. The appropriate model depends on the decision question (Dastgir et al., 2023; Simone et al., 2022; Zhang et al., 2024).

### 2.6.2. Parameter Patterns in Existing Approaches

An important model-design question concerns how many parameters should be included in the generic core and its domain-specific instantiations.

The literature suggests that the appropriate number cannot be fixed in advance; parameter choice follows the semantics and purpose of the model (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

Drainage approaches already illustrate this diversity. Standard centrality may use only graph structure, whereas tailored centrality can incorporate intrinsic node relevance and flow direction (Simone et al., 2022), (Simone, 2023). Dastgir et al. (2023) combine runoff-related path information with capacity-related information for critical-pipe analysis. Meijer et al. (2023) focus on subsystem storage and discharge capacities together with graph-based flow paths. Zhang et al. (Zhang et al., 2025) combine pipeline risk factors, multiple centrality measures, edge vulnerability, and propagation-path information. Hydraulic and learned-surrogate approaches use substantially larger sets of rainfall, boundary, geometric, control, and hydraulic-state variables (Zhang et al., 2024).

These studies show that more parameters are not automatically better, just as fewer parameters are not automatically preferable. Each added variable should represent a distinct mechanism or provide empirically useful domain information. Adding several correlated or weakly justified parameters may reduce interpretability without improving the decision task (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Across the reviewed methods, a defensible reusable core retains only semantically distinct roles while allowing each domain to construct those roles from several observed variables where justified. For example, a drainage susceptibility mapping may combine hydraulic or physical attributes without requiring a generic propagation equation to absorb every raw parameter (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

Thus, the literature supports a generic core plus domain-supported parameterization rather than a universal fixed list of physical variables. The final number of drainage-specific parameters should be determined from evidence, data availability, and validation rather than from an arbitrary target count (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2024).

**Table 2.5.** Parameter patterns in existing approaches

| Example approach | Representative information used | Role in model | Literature-derived implication |
|---|---|---|---|
| Tailored centrality (Simone et al., 2022; Simone, 2023) | Topology, intrinsic relevance, flow direction | Improves structural ranking | Topology can be enriched, but added variables need clear semantics |
| Critical-pipe graph method (Dastgir et al., 2023) | Runoff/path information and capacity-related information | Approximates criticality/flooding consequence | Shows value of separating network paths from capacity response |
| Weakest-link method (Meijer et al., 2023) | Storage, discharge capacity, flow paths | Screens subsystem weakness | Domain variables may be combined without redefining the generic graph concept |

| Example approach | Representative information used | Role in model | Literature-derived implication |
|---|---|---|---|
| Siltation-chain analysis (Zhang et al., 2025) | Pipeline risk, centrality, edge vulnerability, risk path | Ranks critical nodes and paths | Multiple parameters are justified when each represents a distinct mechanism |
| GNN hydraulic surrogate (Zhang et al., 2024) | Hydraulic states, runoff, boundary, control variables, and graph topology | Predicts physical hydraulic states | High parameter count supports richer prediction but changes the task and data burden |

## 2.7. Reusable Architectures and Domain-Specific Instantiation

A model can be generic in formulation without being empirically generalizable to every domain. Network science repeatedly reuses architectural ideas—nodes, directed relations, current state, transmission, response, and aggregation—while redefining their meanings for influence, epidemics, cascading infrastructure, cybersecurity, finance, transportation, and drainage (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Faraji et al., 2026; Petrone and Latora, 2018; Simone et al., 2022).

This distinction is important because the word *generalizable* can imply demonstrated performance on unseen domains. A reusable architecture supports cross-domain instantiation only when each application can defensibly map its data and semantics to the abstract roles and validate the resulting interpretation (Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

The core propagation logic can therefore remain fixed while a domain supplies its own parameter definitions and, where justified, an additional refinement layer. Drainage, cybersecurity, transportation, and supply-chain applications require different parameter definitions, evidence, and independent validation even when they share a graph-based computational pattern (Kavallieratos et al., 2021; Zhou et al., 2023; Petrone and Latora, 2018; Dastgir et al., 2023).

This structure resembles a reusable computational backbone with application-specific adaptation, but it should not be interpreted as machine-learning transfer learning. A learned surrogate transfers fitted representations or parameters; an explicit architecture instead reuses a declared computational rule whose variables must be reinterpreted and validated in each domain (Zhang et al., 2024).

## 2.8. Synthesis of the Research Problem and Evaluation Gap

The closing synthesis connects the evidence on state semantics, mechanisms, ranking, and cost to a bounded comparative-risk design problem. It does not presume that a new model name establishes a new mathematical mechanism.

### 2.8.1. Synthesis of Research Problems

The reviewed evidence supports three major research problems.

First, local condition does not by itself represent network-context priority. Local monitoring can quantify condition at a component, but comparative prioritization in an interconnected system requires consideration of propagated network effects (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023).

Second, structural importance does not by itself represent current operating response. Centrality and topology use the network as context, but drainage and broader infrastructure studies repeatedly augment structure with hydraulic relevance, capacity, storage/discharge relationships, vulnerability, or other operating information (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Buldyrev et al., 2010; Faraji et al., 2026).

Third, existing propagation states are designed for different analytical objectives. Activation, probability, infection, physical load, failure, financial loss, and hydraulic state each imply different update and aggregation rules. A bounded continuous non-probabilistic comparative risk state therefore requires an explicitly declared semantic and compatible propagation architecture (Kempe et al., 2003; Kermack and McKendrick, 1927; Sheyner et al., 2002; Buldyrev et al., 2010; Faraji et al., 2026; Zhang et al., 2024).

Together, these problems create a need for a model that can integrate local disturbance, current propagated state, and relation-specific operating response while remaining interpretable enough to explain why one node receives a higher comparative risk state than another (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

### 2.8.2. Unresolved Design and Evaluation Question

The operator-level problem is not the absence of bounded aggregation. Probabilistic sum already supplies it; leaky noisy-OR combines a background term and parent influences; and related risk models propagate current states through modifiers (Klement et al., 2004; Fallet-Fidry et al., 2012; Faraji et al., 2026). The study will not claim invention of these components or infer that their particular packaging proves a first-of-its-kind architecture.

The unresolved question for the intended static comparative-index task is when the selected recursion gives accurate receiving-state rankings under a declared reference process, how shared origins and uncertain local inputs limit that agreement, and what computational and data costs accompany alternative aggregation and dependence treatments. Relation heterogeneity and saturation will remain separate behavior questions. The established algebra and the related models do not themselves determine which aggregator is appropriate for this declared interpretation. The study will examine maximum, probabilistic-sum, and bounded-sum conditions while preserving input and recursion controls. Association-based cascade bounds and restricted-tree approximation already exist (Abbe et al., 2017, 2020; Chen et al., 2010a); the proposed benchmark will investigate bracket tightness and ranking quality, not claim that bounding cascade marginals or pruning influence paths is newly introduced. Independent Cascade provides a conditional-transmission process and live-edge reference

(Kempe et al., 2003); it does not supply calibration for the domain index. A controlled stochastic benchmark can expose dependence effects that agreement with GBBRPM itself cannot measure. The ancestry-overlap diagnostic will remain a candidate whose reference accuracy must be tested, distinct from MIA's restricted-tree calculation and LDAG's Linear Threshold computation.

This is an evaluative research opportunity whose contribution will depend on the specification, controlled comparisons, and interpretation of evidence. It is not a verified absence claim about all earlier risk models. The literature search and equation-level comparison will be extended before any stronger novelty statement is made.

A separate question concerns whether domain mappings support the same index interpretation without changing the core. Comparable variable roles do not create a common empirical risk scale, and coherent instantiation does not establish outcome usefulness. Physical or operational benefit will require independent reference evidence. Computational efficiency, rank changes, or disagreement with a higher-fidelity simulator will not substitute for that evidence.

Figure 2.1 connects these precedents to the specific evaluation question rather than suggesting that bounded aggregation has no prior art.

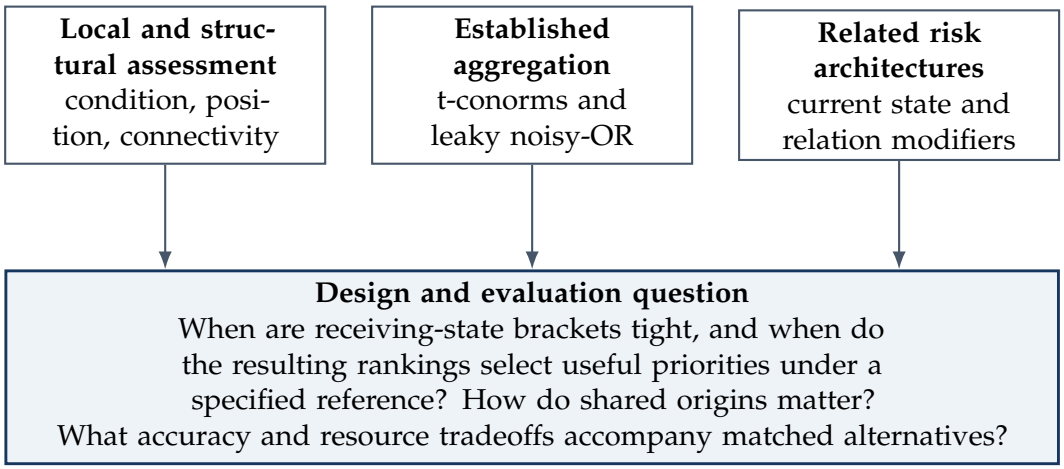



**Figure 2.1.** Literature-derived precedents and the proposed evaluation question. The question concerns selection and behavior of established mechanisms, not invention of bounded aggregation.

### 2.8.3. Chapter Synthesis

The literature connects local condition assessment, structural prioritization, and domain-rich physical simulation with established propagation mechanisms. It also supplies bounded non-probabilistic aggregation and close risk architectures (Klement et al., 2004; Fallet-Fidry et al., 2012; Faraji et al., 2026). These precedents require an explicit account of adopted mathematics rather than a novelty argument based on a new acronym or output label.

The study's methodological response will specify the intended index and investigate receiving-state reference accuracy, matched aggregation and recursion alternatives, shared-origin dependence, and computational cost. A separate stochastic benchmark will make those comparisons falsifiable without

changing the domain index semantics. Relation heterogeneity and robustness will supply complementary behavior evidence. Domain instantiation will be reported separately from physical-outcome validation. Chapter 3 will describe that specification and evaluation plan; the contribution will be assessed through the evidence produced rather than presumed from the model name.

# CHAPTER III
# METHODOLOGY

## 3.1. Research Design and Study Phases

The study will follow a quantitative model-design and computational-evaluation approach. It will formulate GBBRPM, implement the formulation in a deterministic evaluation engine, prepare documented inputs, and assess the resulting behavior and resource requirements. Its scientific emphasis will be the receiving-state accuracy–cost tradeoff under a separately defined stochastic reference, with controlled shared-origin dependence and task-matched comparisons. Three separate domain cases will examine coherent instantiation of the same equations. An interactive prototype will expose the engine's inputs, calculations, and comparative outputs.

The research will proceed through design, implementation, data preparation, evaluation, and interpretation. Literature synthesis will inform the design requirements rather than stand as a separate empirical-validation objective. The model specification and evaluation protocol will be versioned before the planned evaluation is executed. Implementation checks may reveal defects that require correction; any resulting specification or protocol change will be recorded and affected tests will be repeated.

**Table 3.1.** Planned objective–method–evidence alignment.

| Objective | Planned method | Expected evidence or output |
|---|---|---|
| 1: Formulate | Requirements, equations, assumptions, analytical arguments | Specified GBBRPM architecture and properties |
| 2: Implement | Deterministic engine, schema checks, prototype integration | Versioned evaluation engine and traceable prototype |
| 3: Compare | Exact/Monte Carlo references, matched aggregators and recursion ablation | Receiving-state errors, fixed-budget selection, and separate input effects |
| 4: Assess dependence | Matched shared/independent sources, diagnostic evaluation, property and robustness checks | Dependence conditions, diagnostic gains/failures, and core properties |
| 5: Characterize cost | Complexity, matched resource measurements, reference precision, input inventory | Computational and data-requirement account of lightweight design |
| 6: Instantiate | Separate DAG construction, mappings, and within-domain analyses | Case-specific feasibility and applicability boundaries |

Preliminary implementation and exploratory experiments have already been conducted. Their selected outputs are documented in Appendix D. They

will inform feasibility and limitations; they will not be described as unseen confirmatory evidence. The planned study will finalize and freeze its revised specification and protocol before the next evaluation stage.

## 3.2. Design Requirements and Model Boundary

The formulation will address six requirements: a bounded non-probabilistic node state; simultaneous representation of local and inherited disturbance; direction-constrained influence; relation-specific modification; deterministic DAG evaluation; and traceable inputs and outputs. Existing propagation literature motivates separating the meaning of the propagated state from the mechanisms used to update it (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Muñoz-González et al., 2017; Petrone and Latora, 2018).

The study will distinguish GBBRPM, the named mathematical model, from the **GBBRPM evaluation engine**, its software implementation. The core will specify how supplied bounded inputs are combined. The primary output will rank accumulated receiving state. Inspection selection will be evaluated only against a stated outcome, and intervention value will require a separate before-and-after task. The stochastic reference will be an evaluation model with its own primitive assumptions, not a replacement domain adapter. Domain adapters will supply their meanings and calculations; an adapter will not be permitted to silently replace the core update rule.

A domain case will be eligible only if it can provide defensible node and edge meanings and a valid DAG. A directed graph is not necessarily acyclic. Cyclic or bidirectional physical systems will not be declared compatible merely because an importer produces directed records. Any extracted subgraph, consolidation, or excluded edge will be documented together with its effect on the interpretation boundary. Modeling feedback would require a separately specified extension.

## 3.3. Detailed Formulation of GBBRPM

### 3.3.1. Graph, State, and Input Contract

The proposed graph is

$$G = (V, E), \qquad G \text{ is a directed acyclic graph.} \tag{*Equation 3-1*}$$

Nodes will represent domain entities, and an edge $(i, j)$ will represent permitted influence from $i$ to $j$. The predecessor set is $P(j) = \{i : (i, j) \in E\}$. Direction means modeled influence, which must be justified for each instantiation; it need not be physical flow in every domain.

**Table 3.2.** Shared notation and supplied versus derived quantities.

| Symbol | Role and admissible domain | Status |
|---|---|---|
| $G = (V, E)$ | Directed acyclic entity–relation graph | Supplied and validated |
| $B_i$ | Local disturbance, $0 \leq B_i \leq 1$ | Supplied or adapter-derived |
| $S_{ij}$ | Relation-specific susceptibility, $0 \leq S_{ij} \leq 1$ | Supplied or adapter-derived |
| $\tau_{ij}$ | Transmission modifier, $0 \leq \tau_{ij} \leq 1$ | Supplied; neutral baseline 1 |
| $Q_{ij}$ | Bounded incoming contribution | Derived by the engine |
| $R_i$ | Dimensionless comparative risk state, $0 \leq R_i \leq 1$ | Derived by the engine |
| $L_{ij}, C_{ij}$ | Compatible loading and positive capacity | Optional domain-adapter primitives |

$R_i$ will be a continuous comparative index, not a calibrated probability. Its interpretation will be conditional on the selected graph, scenario, and normalization. Values from different domains will not be placed on a common severity or likelihood scale merely because they lie in $[0, 1]$.

$B_i$ will represent disturbance introduced directly at node $i$. $S_{ij}$ will describe the response of a relation or receiving context. $\tau_{ij}$ will describe transfer along that relation. The baseline will set $\tau_{ij} = 1$ where independently defensible heterogeneous values are unavailable. A non-neutral uniform sweep will test attenuation, not empirical calibration.

### 3.3.2. Relation-Level Contribution and Identifiability

For a directed edge, the model will use

$$Q_{ij} = S_{ij}\tau_{ij}R_i. \qquad \text{(Equation 3-2)}$$

This specified rule will make the contribution zero when the upstream state or either modifier is zero and will ensure that the contribution cannot exceed the current upstream state. It is a bounded response construction, not a physical law of transported load or energy. Branching will allow the same upstream state to influence several receivers without interpreting it as a conserved quantity that must be divided among outgoing edges.

The effective edge weight is $w_{ij} = S_{ij}\tau_{ij}$. For fixed topology and local disturbances, any admissible factor pairs with the same product give the same node outputs. Separate interpretation of susceptibility and transmission will therefore require independent measurements or justified constructions; risk outputs alone do not identify both factors.

### 3.3.3. Aggregation Design and Its Rationale

The complement of the receiving-node state is specified as

$$c_j = (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}), \qquad R_j = 1 - c_j. \qquad \text{(Equation 3-3)}$$

Thus the proposed update is

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}). \qquad \text{(Equation 3-4)}$$

The local complement and incoming complements jointly determine the remaining modeled margin. Equation (3-4) is exactly the associative extension of the established probabilistic-sum t-conorm to the local input and all incoming contributions (Klement et al., 2004). The name GBBRPM will identify the specified graph-index architecture rather than a new aggregation operator. The chosen non-probabilistic interpretation does not confer probabilistic independence or event-union semantics, and does not itself establish a new mathematical mechanism.

The study will explain this choice relative to alternative bounded aggregators. Sum with clipping admits cumulative contributions but reaches its cap directly; a maximum retains only the strongest input; and an arithmetic mean can fall when a weak input is added. The selected product complement permits several nonzero contributions to increase the state without exceeding one. This rationale concerns required mathematical behavior; it does not establish superior physical realism. The planned matched-aggregation comparison in Section 3.6.1 will assess that selection explicitly. RIPA's current-state modifiers and leaky noisy-OR's background-plus-parent form will be compared as close prior art (Faraji et al., 2026; Fallet-Fidry et al., 2012); their domain-specific probability or risk meanings will not be silently replaced to claim a matched performance benchmark.

### 3.3.4. Initialization, Recursion, and Output

For a node with no predecessors, the empty product is one, giving

$$P(j) = \emptyset \quad \Longrightarrow \quad R_j = B_j. \qquad \text{(Equation 3-5)}$$

A source node may therefore contain local disturbance. Its risk is not automatically one. An isolated node follows the same rule.

Nodes will be evaluated in a topological ordering. All predecessor states will be available before a receiver is computed. Each newly computed state will immediately become the upstream state for later relations; the engine will not substitute the original $B_i$ for an accumulated $R_i$. Disconnected components will be evaluated independently.

Figure 3.1 illustrates these mechanisms. Appendix A provides a complete controlled calculation. The schematic and calculation explain the model; they are not independent domain validation.

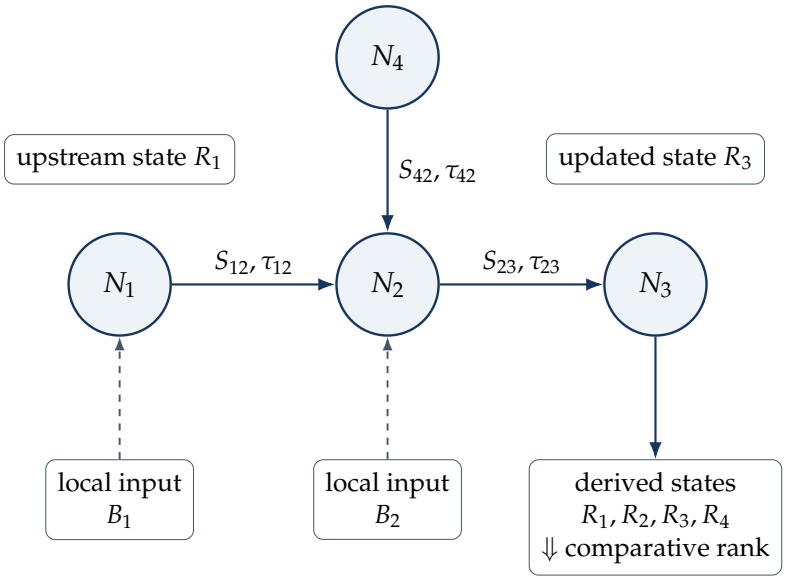



**Figure 3.1.** Generic directed-node example showing local inputs, relation-level modifiers, recursive state propagation, convergence at $N_2$, and ranking from the computed node states.

Final $R_i$ values will be sorted in descending order. Ties will represent equal modeled standing under the declared numerical policy rather than receive an unsupported decision priority. Ranking is a derived output, not an additional propagation mechanism.

### 3.3.5. Analytical Properties and Dependence Boundary

The operator's basic boundedness and monotonicity are established properties (Klement et al., 2004). The following arguments will confirm their consequences for this specified recursion, not claim a new theorem about t-conorms. Boundedness follows because every term in Equation (3-4) belongs to $[0, 1]$. For an incoming contribution $x_k = Q_{kj}$,

$$\frac{\partial R_j}{\partial x_k} = (1 - B_j) \prod_{i \neq k} (1 - x_i) \geq 0, \qquad \frac{\partial R_j}{\partial B_j} = \prod_i (1 - x_i) \geq 0. \qquad \textit{(Equation 3-6)}$$

These expressions establish local monotonicity. On a fixed DAG, global monotonicity with respect to independently increased $B$, $S$, or $\tau$ follows by induction over topological order. This argument assumes the other primitive inputs and parameter-construction rules are held fixed; it does not guarantee unchanged rankings.

If $B_j = 1$ or any incoming $Q_{ij} = 1$, the receiving state is one. A zero modifier gates its edge. By contrast, $S_{ij} = 1$ alone leaves $Q_{ij} = \tau_{ij} R_i$ and does not guarantee $R_j = 1$. The clipped $L/C$ adapter can saturate an edge's susceptibility before the network's states saturate.

When one disturbance splits and reconverges, incoming contributions can share ancestry. Equation (3-4) will still combine them separately. Boundedness therefore does not establish dependence correction. This limitation will be evaluated explicitly rather than hidden by a stable ranking or a high receiving-node state.

## 3.4. Evaluation Engine and Prototype Design

### 3.4.1. Per-Scenario Engine Algorithm

The engine will use node records and directed adjacency lists. For each scenario it will perform the following steps:

1. Validate unique node identifiers, edge endpoints, parameter types, finite values, admissible bounds, and the declared input representation.

2. Reject self-loops, duplicate directed edges, and cyclic graphs. Generate and retain a topological order for valid input.

3. Obtain $S_{ij}$ directly or derive it through the declared adapter. Set neutral $\tau_{ij} = 1$ only where the contract permits an omitted value.

4. For each node in topological order, compute all incoming $Q_{ij}$, apply Equation (3-4), and retain the node and contribution trace.

5. Rank the resulting node states with the declared tie policy and export outputs together with scenario and engine-version metadata.

The interface will not infer missing disturbances from graph position. Node $B$ and edge $S$ or its complete primitive pair will be required. An explicit zero disturbance will be distinguished from an unavailable observation. If both direct $S$ and an $L/C$ pair are provided for an edge, the final contract will reject the ambiguous representation. Capacity must be positive, load nonnegative, and units compatible.

Input violations will produce a reasoned validation failure instead of silently clipping malformed inputs. Clipping will apply only to the declared susceptibility mapping. Floating-point behavior near zero and saturation will be checked against high-precision worked cases. The engine will report the numerical tolerance used by its tests and ranking policy; any change to the arithmetic implementation will require regression comparison with the specified equations.

### 3.4.2. Prototype Integration and Visualization

The research prototype will connect a React/Vite graph interface, a FastAPI validation service, and the versioned GBBRPM evaluation engine (Figure 3.2). Synthetic and user-imported DAGs will use the same input contract; an operational adapter will remain inactive until suitable provenance-complete data are available. The evaluated package version or source revision will be recorded.

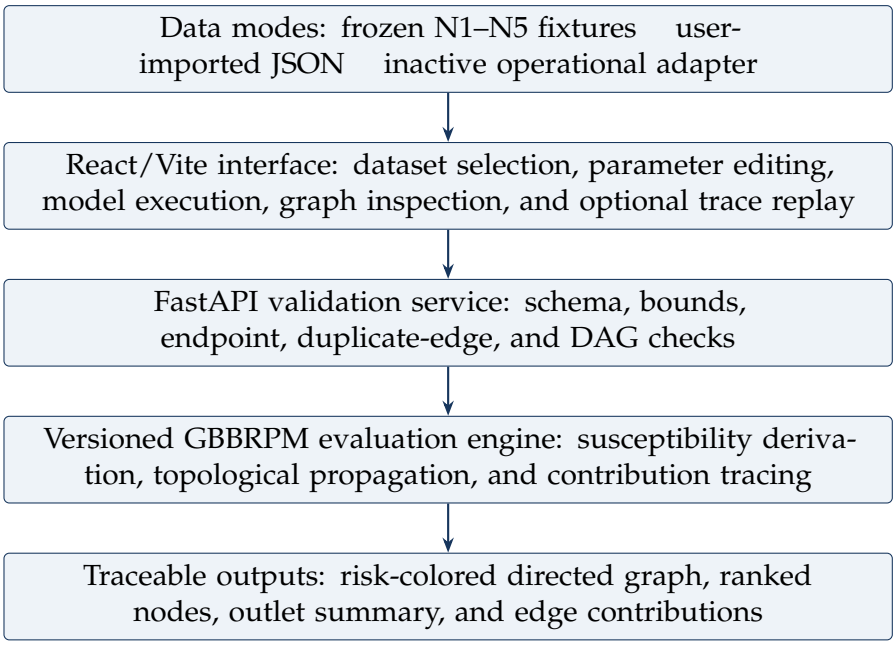



**Figure 3.2.** Proposed research prototype architecture separating data modes, interface, validation service, GBBRPM evaluation engine, and comparative outputs.

The interface will expose validated inputs, node rankings, and engine-computed $B$, $Q$, and $R$ traces. Optional replay will present stored results, with concurrent independent branches and receiving states that settle after incoming contributions arrive. Display modes will preserve numerical results; their timing will not represent physical propagation speed. Trace assembly and rendering costs will be distinguished from scalar evaluation.

Prototype work for this study will prioritize functional consistency and inspection of the specified model. Additional presentation features will be outside the planned mechanism evaluation. Appendix E records the supporting contract and outstanding checks.

## 3.5. Data Sources, DAG Construction, and Parameterization

Each domain will be documented using the same template: source and version; node and edge meanings; direction and acyclicity; local disturbance; susceptibility and transmission; missingness or excluded records; and permitted output interpretation. Observed, documented, derived, assumed, synthetic, and simulated fields will be labeled separately.

### 3.5.1. Drainage Instantiation

Nodes and directed relations will represent the chosen drainage entities and permitted downstream influence. A load-to-capacity adapter will use

$$u_{ij} = \frac{L_{ij}}{C_{ij}}, \qquad S_{ij} = \min(1, u_{ij}), \quad L_{ij} \geq 0,\ C_{ij} > 0. \tag{Equation 3-7}$$

Local $B_i$ values will represent the declared disturbance scenario or documented condition mapping. Synthetic inputs will not be called operational measurements. Operational records will be sought for topology, flow direction, specifications, loading/capacity, inspection and maintenance history, and relevant outcomes. Their adequacy will be assessed before any field-validation claim is made.

SWMM scenarios will supply an external hydraulic reference. They will be aligned to the corresponding graph, intervention, and node mapping. Acyclic downstream influence does not represent hydraulic backwater, surcharge, ponding, or redistribution. Simulated reference outputs will remain distinct from field observations.

### 3.5.2. Software-Dependency Instantiation

Packages will represent nodes, with edges directed from a dependency to its dependent so that modeled influence reaches the consuming package. The preliminary Express 4.18.2 case has 71 packages and 128 edges. Version-specific advisory evidence and the declared severity mapping will provide $B_i$; severity categories will be treated as bounded index inputs rather than calibrated exploit probabilities.

The initial case will retain $S_{ij} = 1$ and neutral baseline transmission. The uniform sensitivity sweep will use $\tau \in \{0, 0.25, 0.50, 0.75, 1.00\}$. The topology and advisory snapshots will be versioned. Cyclic dependency input will be rejected by the DAG engine; any separately proposed component-condensation approach would require its own node semantics.

Shared `qs` and `send` paths in the preliminary Express case will be identified as reconvergence rather than independent sources. The case will test mapping and parameter response, not future incident prediction.

### 3.5.3. Electrical Instantiation

The electrical workflow will separate full-network import verification from the observable Component 2 mechanism study using the authorized Caltech SoCal 28-Bus data (Xie et al., 2025). The preliminary full import contains 187 bus records and 177 active connections; import counts alone will not establish physical acyclicity or fault prediction.

Component 2 will be selected using observability, authorized topology, traceable meters, and DAG compatibility. Zero-impedance-connected buses will be consolidated using the documented adapter. The preliminary complete component has 13 nodes and 12 elements; exclusion of an unobserved element leaves a 12-node, 11-edge arborescence. This fit-for-purpose case will not be treated as a representative sample of distribution grids.

Synchronized phasors will provide real and reactive loading. For $T$ aligned snapshots, downstream powers will first be aggregated within each snapshot, then converted to apparent power:

$$L_{ij} = \frac{1}{T} \sum_{t=1}^{T} \sqrt{P_{ij,t}^2 + Q_{ij,t}^2}. \qquad \text{(Equation 3-8)}$$

The magnitude of mean $P$ and $Q$ will not be substituted for this mean of magnitudes. Transformer capacity will follow the documented nameplate policy; three-phase line capacity in kVA will be $3V_{\text{phase}}I_{\text{rated}}/1000$ for voltage in volts and current in amperes. Equation (3-7) will then supply electrical susceptibility.

The source and terminal-meter assignments, synchronization tolerance, unusable meters, neighboring-feeder exclusion, and unallocated residual power will be recorded. The preliminary calculation used 24 normal-operation snapshots. Declared disturbances will be injected for controlled scenarios; they will not be relabeled as observed faults. Mains-register unit inconsistencies will be documented, with phasor-derived loading retained as the selected input.

Independent switching-event analysis will be conditional on event-aligned measurements and outcomes. Existing normal-operation coverage does not provide those outcomes. Restricted raw measurements and credentials will not be included in the public manuscript package.

### 3.6. Controlled Experiments and Comparative Conditions

The research will use linear, branching, converging, reconverging, and mixed DAGs to separate mechanisms before examining combined behavior. Figure 3.3 is a schematic overview; the executable fixtures, not the abbreviated diagram, will define every node and edge.

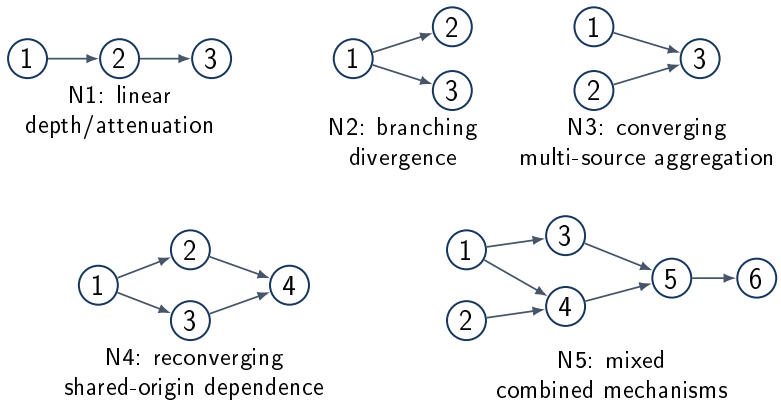



**Figure 3.3.** Schematic examples of the controlled DAG structures. Arrows indicate permitted influence direction; these abbreviated diagrams do not show every node in the executable N1–N5 fixtures.

The preliminary N1–N5 inventory contains 6, 8, 8, 7, and 20 nodes and 5, 7, 7, 7, and 27 edges respectively. The reconstructed N1–N5 fixture inputs will be versioned and frozen for reviewed reruns. Their provenance will distinguish them from the earlier fixture inputs, which are unavailable for direct file comparison. Agreement with preserved historical outputs does not establish that the reconstructed files are identical to those earlier inputs.

**Table 3.3.** Preliminary scenario configuration proposed for reviewed reruns.

| Family | Configuration | Runs |
|---|---|---|
| Source severity | Four levels across five networks | 20 |
| Drainage $L/C$ stress | Five load scales across five networks | 25 |
| Disturbance location | Selected historical locations | 36 |
| Multi-source configurations | Selected source subsets in N3 and N5 | 10 |
| Intermediate disturbance | Three nodes and three severities across five networks | 45 |
| Total | Existing configuration; added protocols counted separately | 136 |

The reviewed reruns will retain severity levels $B \in \{0.25, 0.50, 0.75, 1.00\}$ and load scales $\{0.50, 0.75, 1.00, 1.25, 1.50\}$ unless a documented protocol revision is needed. Location tests will set the selected source to 0.75 while clearing other local disturbances. Intermediate tests will retain baseline disturbances and add the selected local term. Exact fixtures and source subsets will be listed in the run manifest.

Paired comparison conditions will include local-only $R_j = B_j$; the same propagation rule with uniform $S = 0.25$, 0.50, or 0.75; and heterogeneous-susceptibility GBBRPM. A topology-only descriptor will use normalized directed in-degree $d_{\text{in}}(j)/(|V| - 1)$ for $|V| > 1$, with zero for a single-node graph. This descriptor represents incoming structural position, not a physical risk reference.

Changing $B$ or $S$ on fixed topology leaves the structural descriptor unchanged. The analysis will quantify how GBBRPM responds instead. Comparison with a topology score will establish different information dependence, not universal superiority over structural methods. Additional structural-comparison runs will be recorded separately from the preliminary 136-run count.

### 3.6.1. Matched Bounded-Aggregation Comparison

The study will separately compare the maximum, probabilistic-sum, and bounded-sum t-conorms (Klement et al., 2004). For a bounded input vector $\mathbf{x}$, their definitions are

$$\mathcal{A}_M(\mathbf{x}) = \max_k x_k, \qquad \mathcal{A}_P(\mathbf{x}) = 1 - \prod_k (1 - x_k),$$
$$\mathcal{A}_L(\mathbf{x}) = \min\left(1, \sum_k x_k\right). \qquad \text{(Equation 3-9)}$$

The input vector at each receiver will contain its local $B_j$ and every incoming contribution. Graph, $B$, $S$, $\tau$, and topological order will be held fixed across conditions. Two complementary analyses will be performed: an operator-only comparison using identical supplied contribution vectors, and an end-to-end comparison in which each variant recomputes all upstream states recursively with its own aggregator. This prevents a comparison variant from incorrectly inheriting states computed by the GBBRPM baseline.

For illustration, inputs $(0, 0.5, 0.5)$ give 0.5, 0.75, and 1.0 under maximum, probabilistic sum, and bounded sum respectively. These are algebraic examples,

not new experimental findings. A chain with one source disturbance and no intermediate local disturbance will provide an agreement control because each receiver has only one nonzero input. Independent convergence, multiple local-plus-incoming terms, and reconvergence will expose differing accumulation behavior.

The comparison will use the documented N1–N5 configurations and their severity and source-condition sweeps. Per-node paired differences, terminal states, saturation fraction, Spearman agreement, and tie-aware top-$k$ overlap will be reported. Saturation fraction will count nodes with $R \geq 1 - 10^{-12}$ and will be labeled as a numerical reporting criterion. Constant-rank cases will follow the undefined correlation policy in Section 3.11. Source count and reconvergence will be reported separately.

All three operators are bounded and monotone; those properties alone will not identify a preferred choice. The analysis will characterize accumulation, saturation, and ranking tradeoffs under the declared index semantics. Improved physical or operational prioritization will require an independent reference. Diagnostic variants will be separately versioned in the experiment runner and counted outside the historical 136-run suite; they will not overwrite the frozen GBBRPM baseline.

### 3.6.2. Separately Defined Stochastic Reference

The principal controlled evaluation will examine receiving-state accuracy under a specified stochastic process, alongside the index-behavior analysis. This reference will adopt Independent Cascade's conditional edge-transmission mechanism and live-edge interpretation (Kempe et al., 2003). The study will add independently sampled initial activations at all nodes. For this benchmark only, the supplied numbers $B_j$ and $w_{ij} = S_{ij}\tau_{ij}$ will parameterize independent Bernoulli variables:

$$Z_j \sim \text{Bernoulli}(B_j), \qquad E_{ij} \sim \text{Bernoulli}(w_{ij}), \qquad X_j = Z_j \vee \bigvee_{i \in P(j)} (X_i \wedge E_{ij}). \tag{Equation 3-10}$$

All primitive $Z$ and $E$ variables will be mutually independent; derived node activations need not be independent. The Boolean recursion will be evaluated in topological order. A trial will draw each primitive once and reuse its value wherever its consequences recur. The reference output will be $\pi_j = \Pr(X_j = 1)$, not expected spread obtained by selecting $j$ as a seed.

Conditional transmission will use $w_{ij}$, not $Q_{ij} = w_{ij}R_i$. Sampling the upstream activation and then using $Q_{ij}$ would incorporate the upstream marginal twice. GBBRPM will retain its deterministic index definition. Differences $R_j - \pi_j$ will be called approximation errors only within this explicitly probabilistic benchmark. Agreement will not calibrate the domain adapters or establish real-world failure, exploitation, or inspection utility.

For small fixtures with at most 16 non-degenerate primitive variables, the exact solver will enumerate their binary configurations, evaluate the Boolean recursion, and sum configuration probabilities for each activated node. Probabilities of zero or one will be fixed, not enumerated. The solver will check

total probability mass and chain, isolated-node, independent- convergence, and diamond identities before larger runs proceed. On graphs whose incoming activation events have disjoint primitive ancestry, the complement-product recursion will agree with this reference by independence and induction. The study will state that sufficient condition explicitly; structural overlap will not be assumed to make it necessary.

For larger graphs, the Monte Carlo solver will estimate $\widehat{\pi}_j = N^{-1}\sum_{t=1}^{N} X_j^{(t)}$ from $N = 100{,}000$ independent trials in memory-bounded batches. A predeclared seed will be stored per graph and scenario. Sampling uncertainty will be reported using simultaneous Hoeffding intervals (Hoeffding, 1963) $[\widehat{\pi}_j - h, \widehat{\pi}_j + h]$ clipped to $[0, 1]$, where a two-sided concentration bound and a union bound over nodes give

$$h = \sqrt{\frac{\log(2|V|/0.05)}{2N}}. \qquad \text{(Equation 3-11)}$$

This conservative concentration bound applies to the sampling estimator across nodes in one scenario; it is not an error bound for GBBRPM or a simultaneous guarantee across every experiment. Monte Carlo will be checked against enumeration on the eligible fixtures. Validation discrepancies and interval coverage will be recorded, not silently removed. Graphs sharing primitive draws will share the same reference for all deterministic methods.

### 3.6.3. Benchmark Bracket and Its Interpretation

*Proposition (receiving-state bracket under independent primitives).* Let $G$ be a finite DAG and let all initial-activation and edge-transmission variables in Equation (3-10) be mutually independent, with supplied parameters in $[0, 1]$. On the same graph and inputs, define recursive maximum and clipped-additive states by

$$M_j = \max\left(B_j, \max_{i \in P(j)} w_{ij} M_i\right), \qquad H_j = \min\left(1, B_j + \sum_{i \in P(j)} w_{ij} H_i\right), \qquad \text{(Equation 3-12)}$$

where the incoming maximum is zero for an empty predecessor set. Let $R_j$ be ordinary GBBRPM and $\pi_j = \Pr(X_j = 1)$ under the benchmark. Let $T_j$ be the exact activation probability on the target-specific restricted in-arborescence defined in Section 3.6.5. It will retain a consistent maximum-weight path from every positive-path ancestor, the original local inputs and weights, and target $j$ itself; no path-weight threshold will be applied. Then

$$M_j \le T_j \le \pi_j \le R_j \le H_j \qquad \text{for every } j \in V. \qquad \text{(Equation 3-13)}$$

*Proof.* In a DAG, $X_i$ cannot depend on the primitive $E_{ij}$, and no incoming activation at $j$ can depend on $Z_j$. Independence of the primitives therefore gives $\Pr(X_i \wedge E_{ij}) = w_{ij}\pi_i$. Since activation at $j$ includes its own initial event and every successful incoming event, $\pi_j \ge \max(B_j, \max_i w_{ij}\pi_i)$. A predecessor-free node

has $M_j = \pi_j = B_j$; induction over topological order establishes the lower bound.

The maximum recursion also has the path representation $M_j = \max_{a,\gamma:a\to j} B_a \prod_{e\in\gamma} w_e$, including the zero-length path at $j$. The restricted tree retains a maximum-weight path for each positive-path ancestor by consistent next-edge choices. Activation at $a$ through its own local event followed by successful transmission along that retained path is sufficient to activate $j$ in the tree. Its probability is $B_a \prod_{e\in\gamma} w_e$ by primitive independence. The tree probability is at least each such path contribution and $B_j$, so $M_j \le T_j$. Ancestors without a positive path contribute zero to $M_j$. Couple the restricted and full processes using the same local and edge draws. Deleting edges cannot create an activation, hence $T_j \le \pi_j$. On the target-directed tree, incoming ancestor subtrees have disjoint primitives; complement-product recursion therefore evaluates $T_j$ exactly. This argument relies on the stated unthresholded, path-preserving construction; arbitrary pruning need not retain the $T_j \ge M_j$ guarantee.

For the upper bound, let $F_{ij}$ denote failure of incoming activation through $(i, j)$, that is, the complement of $\{X_i = 1, E_{ij} = 1\}$. Every $X_i$ is an increasing Boolean function of the primitive variables, so every $F_{ij}$ is decreasing. Positive association under a product Bernoulli measure, the Harris/FKG principle, gives $\Pr(\bigcap_i F_{ij}) \ge \prod_i \Pr(F_{ij})$. This established principle is also used for cascade upper bounds by Abbe et al. (2017, 2020); it is not a new correlation inequality. Since $Z_j$ is independent of the incoming events,

$$\pi_j = 1 - (1 - B_j) \Pr\left(\bigcap_{i\in P(j)} F_{ij}\right) \le 1 - (1 - B_j) \prod_{i\in P(j)} (1 - w_{ij}\pi_i). \quad \textit{(Equation 3-14)}$$

Complement product is increasing in each predecessor marginal. Starting from $\pi_j = R_j = B_j$ at predecessor-free nodes, topological induction yields $\pi_j \le R_j$.

Finally, for nonnegative inputs in $[0, 1]$, $1 - \prod_k (1 - x_k) \le \min(1, \sum_k x_k)$. Applying this inequality to the local term and incoming contributions, followed by topological induction and monotonicity of clipped addition, gives $R_j \le H_j$. Primitive probabilities of zero or one follow directly or by continuity. This completes the argument.

The proposition will be presented as a specialization of established association and bounded-aggregation arguments, not as a verified first theorem about cascade approximation. Abbe et al.'s journal treatment also bounds individual activation probabilities before summing them into seed-set influence. Its upper method uses length-indexed messages; its DAG-based lower method uses a different incoming-message recurrence (Abbe et al., 2020). Section 2.4.6 compares the constructions and limits. Independent local activation can also be represented by one certainly active auxiliary source with independent edges of probabilities $B_j$ to all original nodes. That construction facilitates comparison with fixed-seed literature; it will not conceal the model's mathematical precedents.

For this benchmark, score discrepancy is one-sided: $R_j - \pi_j \ge 0$. Its scalar certificate is $0 \le R_j - \pi_j \le R_j - M_j$. When the tree comparator is computed, the tighter certificate is $0 \le R_j - \pi_j \le R_j - T_j \le R_j - M_j$. Computing $T_j$ requires tree construction and evaluation for each requested target; the tighter certificate is not free relative to the scalar pass. Its reporting will reuse the already computed comparator result. GBBRPM is no looser than recursive clipped addition as a

nodewise upper bound; this ordering need not be rediscovered experimentally. Experiments will measure bound width, actual slack, conditions of equality, and ranking or selection effects. A smaller pointwise score error does not guarantee better rank order. The condition $T_a > R_b$ certifies $\pi_a > \pi_b$ when tree bounds are available; $M_a > R_b$ remains a cheaper sufficient certificate. Overlapping brackets remain inconclusive, not evidence of a reversed ordering.

These guarantees concern only the stated independent-primitive benchmark. They do not certify probabilities or physical bounds for the domain index, dependent initial events, correlated transmission primitives, feedback, or other cascade mechanisms. The ancestry-aware diagnostic will require its own analysis and will not inherit the GBBRPM upper-bound guarantee.

### 3.6.4. Matched Dependence Controls and Graph Inventory

The first experiment will separate uncertain common origins from independent origins on a fixed graph. A root $a$ will connect with weight one to $m$ branch-start nodes $s_1, \ldots, s_m$. Each branch start will connect to a common receiver $d$ through an edge-disjoint path of $\ell$ edges, all with weight $p$. Intermediate local disturbances and $B_d$ will be zero. In the shared-origin condition, $B_a = b$ and all branch-start local terms will be zero. In the independent-origin condition, $B_a = 0$ and every $B_{s_r} = b$. The graph, weights, and branch-start activation marginals will match; only the joint source construction will change. Both conditions will include the inactive root in their node inventory.

For $u = p^\ell$, this constructed family has exact receiver references

$$\pi_d^{\text{shared}} = b\{1 - (1-u)^m\}, \qquad \pi_d^{\text{independent}} = 1 - (1-bu)^m. \qquad \textit{(Equation 3-15)}$$

GBBRPM will give $1 - (1-bu)^m$ in both conditions. This identity will provide an analytical benchmark, not a claim of a newly invented dependence theory. For two branches, the shared-origin discrepancy is $b(1-b)u^2$. Thus shared ancestry alone does not force discrepancy: $b = 1$ gives agreement in this family. With $b = 0.5$ and $u = 1$, the reference is 0.5 and GBBRPM is 0.75. With $b = 1$ and $u = 0.25$, both are 0.4375; retaining only the strongest branch gives 0.25. These algebraic controls will prevent strongest-path pruning from being assumed correct in advance.

**Table 3.4.** Planned reference-benchmark inventory, separate from historical runs.

| Family | Fixed protocol | Scenarios |
|---|---|---|
| Matched branch family | Four branch counts, two path lengths, seven edge weights, five source probabilities, and two origin conditions | 560 |
| Generated DAGs | Three families, four sizes, 25 graph seeds, six weights, and two initial-activation settings | 3600 |
| Heterogeneous-weight subset | Three families, four sizes, one development and five evaluation seeds, one weight assignment, and two initial settings | 144 |
| Express topology | One verified snapshot, six weights, and two controlled initial-activation settings | 12 |

The branch-family grid will use $m \in \{1, 2, 4, 8\}$, $\ell \in \{2, 4\}$, $p \in \{0.05, 0.10, 0.20, 0.25, 0.50, 0.75, 1\}$, and $b \in \{0.10, 0.25, 0.50, 0.75, 1\}$, under both origin conditions. The closed form will supply exact receiver references at every size, including rare-event regimes; the 16-variable enumeration cap will remain an implementation-validation choice rather than the limit of exact evidence. Conditioning or decision-diagram solvers will be optional future tooling, not required to establish these references.

The generated-DAG stage will use $|V| \in \{20, 50, 100, 250\}$ and three explicit constructions. Each will have $2|V|$ unique edges and a recorded topological ordering:

a. A forward-edge family will sample edges uniformly from all ordered pairs consistent with a seeded random node ordering.

b. A layered family will distribute nodes as evenly as possible over $\lceil \sqrt{|V|} \rceil$ ordered layers, randomize order within each layer, and sample edges uniformly between adjacent layers. The available-pair count will be checked before sampling; no hidden generator fallback will replace an infeasible configuration.

c. A preferential-attachment DAG will start with all ten forward edges among five ordered nodes. Each later node will choose two distinct earlier predecessors without replacement, proportional to their current total degree plus one, and receive edges from them. This supplies exactly $2|V|$ edges. The construction will be called preferential attachment; a scale-free degree law will not be asserted without evidence.

Per family and size, seeds 41–45 will supply five development graphs and 1001–1020 will supply 20 evaluation graphs, yielding 300 distinct topologies. Each topology will retain its edges under $w \in \{0.05, 0.10, 0.20, 0.25, 0.50, 0.75\}$. The first $\lceil 0.10|V| \rceil$ nodes in its ordering will receive either $B = 0.5$ or $B = 1$; other local terms will be zero. These changes will create 3600 scenarios, of which

720 are development and 2880 are evaluation scenarios. A local-origin node need not have in-degree zero. The controlled uniform weights will isolate topology, dependence, and attenuation; heterogeneous N1–N5 inputs will remain a separate information-construction comparison.

A secondary heterogeneous-weight benchmark will use the same three families and four sizes, development seed 41 and evaluation seeds 1001–1005, and both initial settings. This adds 24 development and 120 evaluation scenarios, separate from the 3600 uniform-weight scenarios and paired to the existing $w = 0.20$ cases on those graphs. For each graph, its $2|V|$ edges will be ordered by the SHA-256 digest of the UTF-8 string `family|size|seed|source|destination|hetero-v1`, with literal family names and integer node identifiers from the graph manifest. Digest ties will be resolved by $(\text{source}, \text{destination})$. The first quarter will receive $w = 0.10$, the middle half $w = 0.20$, and the last quarter $w = 0.30$. These sizes make each count integral and fix the mean at 0.20 without choosing weights from outcomes. In this synthetic benchmark $S_{ij} = w_{ij}$ and $\tau_{ij} = 1$; the same independent-primitive reference and ranking metrics will apply. Paired differences will describe sensitivity to effective-weight heterogeneity, not real-domain calibration or identification of the two factors. Five evaluation graphs per family and size limit inference; this secondary subset will not be merged with the primary uniform summaries.

The record will include depth, degree distributions, reachability, source coverage, and the fraction of receivers with overlapping incoming local-origin lineages. Structural overlap will not be equated with statistical dependence strength. Family comparisons will be descriptive, because depth and degree patterns differ even with equal node and edge counts. The fixed branch family will supply the controlled dependence manipulation.

A further 12 scenarios will use the original Express 4.18.2 topology, containing 71 nodes and 128 edges. The original `nodes.csv` and `edges.csv` are available in the experiment repository under `data/software/express_4.18.2/` and are included in this manuscript package. Both files match the SHA-256 hashes recorded in the historical paper-output manifest. The dependency JSON, OSV advisory snapshot, and case manifest are included alongside them. Input verification confirms valid endpoints, dependency-to-dependent edge direction, acyclicity, and the seven advisory-origin identities; it does not constitute execution of the proposed stochastic benchmark.

For each of the six weights above, the seven verified advisory-origin identities will receive controlled initial probabilities of either 0.5 or 1, with other local terms zero. Real advisory severity values will not be interpreted as calibrated probabilities. This will be a stochastic experiment on real topology, not incident validation. The W8 checkpoint will confirm the selected input hashes, source records, and scenario eligibility before protocol freezing. Any unresolved eligibility issue will be documented and may exclude the affected case without delaying the controlled benchmark. Express results will remain separate from generated-graph averages.

At $N = 100{,}000$ trials, small receiving probabilities may be unresolved. The study will therefore preselect 48 precision-audit scenarios: evaluation seed 1001 for every family and size, $w \in \{0.05, 0.10\}$, and both initial-activation set-

tings. Each will receive an additional independent $N = 1{,}000{,}000$-trial reference with its own interval. Both estimates will be retained; this subset will be selected before outcomes are inspected. Coverage statements will remain per solve, not a joint guarantee for all solves. Zero observed activations will not be called zero true probability. Elsewhere unresolved differences will remain unresolved. Batching will cap memory; development timing will determine whether the full prospective inventory is feasible. Any reduction will be documented and frozen before evaluation outcomes, including its resulting scenario counts.

Before evaluation, a computational pilot will use development seed 41 at sizes 20, 100, and 250 in each family. It will time both initial settings at $w = 0.10$ and 0.75, as well as the heterogeneous assignment, with staged 1,000- and 10,000-trial batches; at least one size-250 case will run at the proposed 100,000 trials if memory permits. Pilot records will distinguish graph generation, reference sampling, deterministic methods (including the published DAG lower comparator), target-specific tree construction, and output storage. Hardware, implementation revision, trials per second, edge-trial throughput, and peak memory will be recorded. The generated-DAG reference entails 360 million trial trajectories across graphs with 210 edges on average, or $75.6 \times 10^9$ potential edge-trial checks under straightforward edge-by-edge execution. The 48 additional million-trial audits add $10.08 \times 10^9$ such checks; the 144 heterogeneous scenarios add $3.024 \times 10^9$ at 100,000 trials. These counts are workload estimates, not measured runtimes; implementations may avoid many literal checks. Extrapolations will include the extra per-target tree costs and be compared with the compute allocation recorded before evaluation. If the plan is not feasible, batching or a validated equivalent implementation will be tried first. Any reduced scenario grid, trial count, or precision-audit inventory will be declared with revised interval widths and graph-level replication before held-out outcomes are inspected. The controlled branch family, family/size coverage, and independent evaluation graphs will receive priority over extra display work or exploratory domain expansion.

### 3.6.5. Task-Matched Comparators and Recursion Ablation

Every deterministic method will receive the same graph, $B$, and effective weights $w$. The comparison will include local-only scoring and the three recursive aggregators in Equation (3-9). A recursion ablation will retain the complement-product operator but substitute original local input for current upstream state:

$$R_j^{\text{local-upstream}} = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - w_{ij} B_i). \qquad \textit{(Equation 3-16)}$$

This condition will isolate inherited multihop state rather than be called an equivalent cascade solver. Recursive maximum aggregation will supply the strongest local-origin weighted-path comparator; an additional strongest-path method would duplicate it under these inputs. An arithmetic mean will not be required because adding a zero input can reduce its state, contrary to the selected accumulation requirement.

A MIA-inspired restricted-in-arborescence comparator will also be evaluated (Chen et al., 2010a). For each target $j$, reverse topological dynamic pro-

gramming will compute the maximum positive weighted-path product from every ancestor to $j$. Each ancestor will retain one next edge on a maximizing path, with exact ties broken consistently by node identifier. The union of these target-directed paths will form an in-arborescence. No positive-path threshold will be applied in the primary comparator. Original local $B$ terms on retained nodes and the retained edge weights will be preserved. Complement-product recursion on that tree will compute the target's exact activation probability for the restricted independent-primitive process. Edge deletion makes this a lower bound on the full benchmark marginal. This will be an adaptation of MIA's tree-restriction idea to fixed receiving queries, not its complete greedy seed-selection algorithm. Building one tree per target will have a different cost from a single global scalar pass.

The nodewise lower messages of Abbe et al. (2020, Section 4.1, Equation 22) will provide a published-bound comparator. For this condition only, an always-active auxiliary source will connect to each original node $j$ by an independent edge of probability $B_j$; the original edges retain $w_{ij}$. The resulting fixed-seed live-edge process has the same original node marginals as Equation (3-10). The published minimum-distance directed acyclic subnetwork and lower recursion will be implemented as specified, with its nodewise outputs compared against exact small fixtures and timed separately. It may delete original edges even though the input is already a DAG. No ordering relative to $M_j$ or $T_j$ is assumed. The journal upper method uses length-indexed messages with a different computational cost; its equations will be compared in Section 2.4.6, but the full upper algorithm is not required for the fixed-budget method comparison.

The ancestry-aware diagnostic in Section 3.7 will also be compared with the same reference. Each recursive variant will compute its own upstream states. In-degree will remain a descriptive structural comparison; descendant count and seed-spread measures will not be used as receiving-probability accuracy baselines without a separately specified task. Exact and Monte Carlo methods will be reference solvers, with their precision and computational cost stated.

### 3.6.6. Bound Tightness and Fixed-Budget Ranking Metrics

Fixed-budget selection and ranking will be the primary benchmark outcomes; score evaluation will characterize the analytic bracket's tightness. Per scenario, the study will report scalar width $R_j - M_j$, tree-refined width $R_j - T_j$, actual upper slack $R_j - \pi_j$, both lower slacks $\pi_j - M_j$ and $\pi_j - T_j$, mean and maximum absolute error, and terminal error relative to exact or estimated marginals. The guaranteed nonnegative GBBRPM slack will not be treated as an untested hypothesis. Numerical bound violations against exact references will be investigated as implementation or assumption defects. Node averages will first be computed within a graph; cross-graph summaries will give graphs equal weight and show their spread. Graph families, sizes, weight regimes, and source settings will be reported separately. Each graph will remain the replication unit; its 12 weight/source conditions will not be treated as independent graphs. Evaluation summaries will include paired method differences and their distribution across the 20 graph replicates. Development summaries will remain separate. Monte Carlo intervals will accompany score discrepancies. For example, a Monte Carlo interval $[L_j, U_j]$ can be intersected with the analytic bracket $[T_j, R_j]$ when the tree

comparator is available (or $[M_j, R_j]$ for a scalar-only certificate). If nonempty, $[L'_j, U'_j]$ gives possible upper slack $[R_j - U'_j, R_j - L'_j]$. An empty intersection will be recorded as a sampling-coverage or implementation discrepancy, not silently repaired. A difference smaller than reference uncertainty will not be interpreted as a resolved advantage. Reports will include the fractions of estimated reference states and model scores at or above 0.95, the existing numerical-saturation fraction, and the number of distinct score groups. Near-saturation and near-zero cases will not be discarded because they give inconvenient rankings. Bracket-certified pairwise orderings will be counted separately from Monte Carlo-based rank estimates.

Ranking evaluation will exclude nodes with positive local inputs, so that high supplied disturbances do not trivially dominate a receiving-state task. The common candidate set $C$ will contain all remaining nodes, including unreachable nodes. The study will report tie-aware Spearman association, and selection at budgets $k = \max(1, \lceil f|C| \rceil)$ for $f \in \{0.10, 0.20, 0.30\}$. Exactly $k$ nodes will be selected. Boundary ties will be handled by averaging over 100 seeded random node-order permutations, shared by all methods and unrelated to reference outcomes. Permutations will use seed 140 and the $10^{-12}$ anchored score-tie policy stated in Section 3.11. This policy will differ from the expanded cutoff sets used for descriptive Jaccard comparisons in Section 3.11.

For a selected set $\mathcal{T}_k$, the benchmark captured activation mass and selection regret will be

$$\begin{aligned} \text{Capture}(\mathcal{T}_k) &= \frac{\sum_{j \in \mathcal{T}_k} \pi_j}{\sum_{j \in C} \pi_j}, \\ \text{Regret}_k &= \sum_{j \in \mathcal{T}_k^*} \pi_j - \sum_{j \in \mathcal{T}_k} \pi_j, \end{aligned} \qquad \text{(Equation 3-17)}$$

where $\mathcal{T}_k^*$ maximizes activation mass over $k$ candidates. Capture will be undefined when the denominator is zero; the all-zero case will be reported explicitly. Monte Carlo-based oracle membership and regret will be labeled estimates. A conservative bound for the difference in mass between two selected sets is $h$ times their symmetric-difference size; results unresolved at that precision will remain unresolved. These metrics concern expected node activation, not avoided loss or intervention benefit.

A pairwise analysis will explicitly test whether unequal upper slack reverses receiving-node order. On the same candidate set, a strict reversal will have $R_a > R_b + \epsilon$ but $\pi_a < \pi_b - \epsilon$, with $\epsilon = 10^{-12}$. For Monte Carlo references, an interval-resolved reversal will require $U'_a < L'_b - \epsilon$ using the nonempty intersected intervals above. This classification will be conditional on the per-solve sampling coverage in Equation (3-11), not an unconditional proof of reference order. The reversal rate will use pairs with strict model order and a resolved strict reference order as its denominator. Counts and the fraction of strict-model pairs with resolved reference order will accompany the rate; ties and unresolved pairs will be reported separately. Estimated rank associations will remain distinct from these interval-resolved reversals. Nodewise slack and selection regret will also be examined alongside recorded lineage overlap, without assuming that more reconvergence necessarily means more error. Associations across unmatched

graph families will be descriptive; the matched branch family will provide the controlled dependence comparison.

The main comparisons will be paired by graph and scenario. They will ask when the analytic bounds are tight and whether complement-product recursion improves receiving-state selection over simpler rules or the restricted-tree comparator. Clipped sum will remain an inexpensive comparator and outer upper bound. Its saturation fraction will be measured rather than presumed to make it uninformative. The main bracket-tightness comparison will emphasize $R$ and $T$; their pointwise bound ordering will not imply ranking superiority. The study will also ask whether the dependence diagnostic improves either measure without unacceptable resource growth. Negative, equivalent, and mixed findings will be retained. The study will describe an empirical applicability boundary rather than presume a correction wins or claim universal probabilistic accuracy for the index.

### 3.7. Properties, Sensitivity, Robustness, and Dependence

Analytical arguments will be complemented by implementation tests for zero disturbance, isolated nodes, chains, independent convergence, reconvergence, zero edge modifiers, and saturated local disturbance. Randomized admissible DAG/input cases will test the implementation against the declared properties. Passing finitely many tests will not replace the analytical argument.

Sensitivity tests will vary severity, source location, susceptibility, uniform transmission, intermediate disturbance, and source combinations, holding unrelated inputs fixed. Outputs will include node and terminal states, maximum and mean risk, risk sum, and node counts above an explicit reporting cutoff. A cutoff will be an analytical choice rather than a physical failure threshold.

The planned robustness protocol will use 600 trials at each relative perturbation magnitude $\delta \in \{0.05, 0.10, 0.20\}$ with seed 41. For this protocol, each edge susceptibility will be independently perturbed as $S'_{ij} = \min(1, \max(0, S_{ij}(1 + \epsilon_{ij})))$, where $\epsilon_{ij}$ is uniform on $[-\delta, \delta]$. Topology, disturbances, and transmission will remain fixed. This stated protocol will be versioned; earlier outputs will not be assumed identical if their generation conventions differ.

#### 3.7.1. Specified Ancestry-Aware Diagnostic and Its Evaluation

The prospective diagnostic will assign each node a set of local-origin labels. A positive local term will introduce that node's own label; a positive incoming contribution will carry the union of its predecessor's labels. At each receiver, positive incoming contributions will be vertices in an overlap graph, joined when their label sets intersect. Connected components will define groups, including transitive overlap; within each group only the largest contribution will be retained. The receiver's local term and these group maxima will be combined by complement product. The variant will compute its own upstream states and carry union label sets in topological order. This definition will be versioned prospectively; it will not retroactively authenticate historical outputs whose exact grouping convention differs.

The grouping may merge partially overlapping lineages that also contain

independent origins, and may discard independent edge opportunities from a certain common source. The matched controls in Section 3.6.4 will test these failure cases. Reference error and fixed-budget selection will be reported alongside ordinary GBBRPM and the restricted-tree comparator, not merely the difference between variants. Runtime, peak memory, lineage-set sizes, and pairwise overlap checks will also be recorded. With $K$ origin labels and in-degree $d_j$ at receiver $j$, a straightforward set implementation can require $O(K \sum_j d_j^2)$ overlap work in addition to propagation and label storage; it will not inherit the scalar core's linear-time claim.

The protocol will not promise a successful correction. The diagnostic may improve some regimes and worsen others. Any newly designed extension will require a revised specification, development evaluation, and unused evaluation cases; it will not be tuned on the held-out reference outcomes and then reported as independently validated. Findings about this diagnostic will not be generalized to all dependence-aware methods.

### 3.8. Computational and Input-Cost Assessment

The lightweight claim will be assessed as a resource and representation tradeoff. With adjacency-list storage and constant-cost edge operations, DAG checking, topological ordering, and propagation require

$$O(|V| + |E|), \quad \text{whereas complete comparison sorting adds } O(|V| \log |V|). \tag{Equation 3-18}$$

Graph and stored contribution-trace memory will be $O(|V| + |E|)$. The claim applies to one scenario with prepared inputs. Domain adapters, data retrieval, many repeated scenarios, and interface rendering have additional costs.

Sparse DAG sizes will initially include $|V| \in \{20, 50, 100, 250, 500, 1000\}$ with recorded edge counts, followed by a denser-series stress test at 20, 50, 100, and 250 nodes with $|E| = \lfloor 0.25|V|(|V| - 1)/2 \rfloor$. A seeded random topological ordering will restrict all edges to forward pairs, and unique pairs will be selected up to the declared edge count. Sparse cases will target three edges per node. Graphs and bounded inputs will be fixed by seed 41. Five warm-up runs and 30 measured repetitions per configuration will provide median, interquartile range, and raw timings. Graph validation/order, propagation, ranking, and total engine evaluation will be timed separately, alongside peak memory under a stated measurement boundary. Hardware, operating system, language, library versions, and timer/memory instrumentation will be recorded.

The same prepared inputs will be used for relevant local, uniform, structural, aggregation, restricted-tree, and recursion-ablation implementations. The ancestry-aware variant will be timed separately with label preparation included. Restricted-tree construction and per-target calculation will also be timed, including all requested targets; it will not be compared as if its preprocessing were free. Simpler baselines may remain cheaper. Reference comparisons will state trial count and sampling precision. They will time one full $N = 100{,}000$-trial solve per general-DAG scenario and repeat the solve five times on one

preselected graph per size (forward-edge family, seed 1001, $w = 0.5$, initial $B = 0.5$). Deterministic methods will retain the 30-repetition protocol on those same graphs. Reference preparation, simulation, and ranking will be separated, and batching/vectorization choices will be reported. A one-pass index and repeated probabilistic estimation will not be claimed to provide identical fidelity simply because they share inputs. A required-input inventory will separately list observed and derived quantities, adapter effort, and missingness requirements. A smaller core will not imply easy data acquisition.

Higher-fidelity comparison will document the modeled quantities, omitted mechanisms, required fields, and workflow costs. A quantitative SWMM speed claim will be made only if a separately documented paired benchmark is performed with declared scenarios, simulation duration, solver settings, hardware, and timing boundaries. A hydraulic simulation and a static comparative-index evaluation remain different tasks.

### 3.9. Separate Domain-Case Evaluation

The controlled reference study will be the primary mechanism evaluation. The three domain cases will support documented instantiation and assessment of evidence limits. Before selecting a primary operational validation case, the study will require an independently measured target, compatible DAG and entity mapping, adequate outcome coverage, and a mapping frozen before final outcome inspection. Software will not be selected automatically because its topology is available. If no case meets these conditions, the contribution will remain a controlled methodological evaluation with supporting instantiations, and practical usefulness will remain unestablished.

For an eligible receiving-state or inspection case, the protocol will specify the outcome and time window, candidate population, input provenance, development/evaluation separation, matched comparators, and fixed inspection budgets before testing. An outcome already used to construct $B$, $S$, or $\tau$ will not serve as an independent label. Consequence captured will use the selected domain outcome; activation-mass capture from the synthetic benchmark will not be relabeled as observed loss. Intervention evaluation would require independently assessed avoided consequence and will not be inferred from current receiving states.

The drainage case will compare scenario-induced priority changes with SWMM maximum-depth and flooding changes under the aligned intervention protocol. Spearman association, priority-set overlap, and the share of positive hydraulic worsening outside the downstream region will be reported. Negative or low agreement will inform the limits of the graph index rather than be converted into evidence of validated utility.

The software case will report root and network states, root rank, and ranking agreement across the uniform transmission sweep. Advisory-derived disturbance is an input, not an independent incident label. A high root state may include shared-path influence, whose contribution is not quantified merely by stable ranking.

The observable electrical case will include reruns of the preliminary 43-

scenario configuration: one zero-disturbance reference; four root severity levels; disturbance at each of 12 nodes; five load scales; five transmission levels; 15 pairs of internal sources; and one all-internal case. Node and terminal risk, risk sums, and affected-node counts will be reported. The arborescence tests local-plus-incoming aggregation and branch coverage, not downstream reconvergence.

Cross-case synthesis will examine defensible mappings, DAG eligibility, core-rule consistency, computational requirements, and evidence limits. Domain scores, correlations, and physical outcomes will not be pooled as interchangeable performance measures. A common equation does not establish common calibration.

## 3.10. Prototype Functional Verification

Functional verification will compare the recorded prototype revision with the versioned engine on the N1–N5 fixtures, valid imports, and malformed inputs. Checks will cover reference states within declared tolerance, trace coverage and deterministic values, receiving-state reconstruction, and controlled rejection of invalid graphs or non-finite numeric inputs.

Replay checks will assess predecessor-settlement order, concurrent branches, input non-mutation, agreement of settled readouts with returned quantities, and invariance of numerical results across display modes. Automated and manual browser checks will inspect the supported controls and graph layout. The detailed checklist and evidence-recording requirements appear in Appendix E. These checks will establish functional consistency in the tested configuration; they will not establish usability, operational utility, or deployment readiness.

## 3.11. Analysis, Evidence Synthesis, and Reproducibility

### 3.11.1. Metrics and Numerical Reporting

Paired scenario differences will be reported at node and network level. Spearman correlation will use average ranks for tied standing; a constant rank vector will be reported as undefined rather than assigned a favorable correlation. Numerical tie grouping will use a declared absolute tolerance of $10^{-12}$, with each descending group's highest score as its anchor.

Top-$k$ sets will use the nominal cutoff $k = \min(3, |V|)$ or $k = \max(1, \lceil 0.20|V| \rceil)$, depending on the declared metric. All scores within the numerical tolerance of the cutoff score will be included. The resulting set can exceed nominal $k$. Jaccard similarity will be $J(A, B) = |A \cap B|/|A \cup B|$. Top-three and top-20% measures will be labeled separately. These descriptive overlap conventions will be applied consistently to planned runs and will not be retroactively attributed to every historical table. Fixed-budget reference evaluation will instead select exactly $k$ nodes using the shared, outcome-independent tie permutations in Section 3.6.6. Expanded cutoff sets will not be used to claim an advantage at a fixed inspection budget.

Boundary and analytical tests will assess formal correctness. Compara-

tor differences without a reference will assess added information dependence; reference-based errors and selection metrics will assess accuracy under the specified stochastic process. Resource measures will assess cost and domain analyses will support only their declared case-specific claims. Output disagreement alone will not establish better prioritization.

### 3.11.2. Provenance and Acquisition Contingencies

The record will include input hashes, source versions, engine revision, scenario settings, random seeds, trial counts, environment, and output files. Reconstructed N1–N5 fixtures will be identified separately from the earlier fixture inputs and will be versioned for reproducible reruns. The original Express node and edge CSVs are available and match the SHA-256 hashes in the historical output manifest. The software topology and electrical measured loading will remain real-data inputs despite the preliminary status of their experiments; input verification does not establish independent operational validation.

The generated-DAG benchmark will use seeds 41–45 for development and 1001–1020 for evaluation after the simulator, diagnostic, metrics, exclusions, and reporting protocol are frozen. The precision-audit subset will also be fixed. Development and evaluation summaries will be reported separately. The analytically specified branch examples will be mechanism controls, not unseen confirmatory discoveries. Any amendment after evaluation outcomes are inspected will be recorded and will require new unused cases for a further confirmatory claim.

Previously examined scenarios will remain exploratory. Any new confirmatory evaluation will specify its criteria and independent data before inspecting outcomes. If suitable agency records or event-aligned electrical evidence are unavailable at the acquisition checkpoint, the study will report that limitation and review the permitted scope with the adviser. Missing field records will not be replaced with invented operational observations.

## 3.12. Timeline of the Study

Figure 3.4 gives a provisional calendar for completing the revised proposal and its planned evaluation. The model formulation, prototype, and exploratory experiments already exist; the opening activity identifies work completed before W1 without assigning its original duration. Their record will be consolidated during proposal revision. Literature review, proposal revision, and data follow-up will continue while the team prepares for the proposal presentation and addresses adviser comments.

The subsequent phases will finalize the specification and engine, validate the exact and Monte Carlo references, implement comparators, and freeze the protocol before held-out evaluation. The W8 data checkpoint will review available domain evidence, mapping completeness, and eligibility of the verified Express snapshot for the planned benchmark. A computational pilot during reference development will assess whether the prospective scenario inventory is feasible. Any reduction will be documented and frozen before evaluation outcomes, as specified in Section 3.6.4. Mechanism comparisons will be followed

by cost and separate domain analyses; prototype checks and writing will overlap these phases. Adviser review and revision buffers will precede and follow the final defense. New evaluation-stage protocol changes will require unused cases under Section 3.11.

The 24-week duration remains a planning placeholder. If W1 begins on October 12, 2026, W24 ends on March 28, 2027, and the W8 checkpoint falls at the start of the week of November 30, 2026. Defense, submission, and event dates will be confirmed separately. Conditional dissemination may extend beyond the core period; its dashed tail is not an additional committed research phase. Data unavailability will narrow the supported domain claims rather than block the independently specified controlled benchmark.

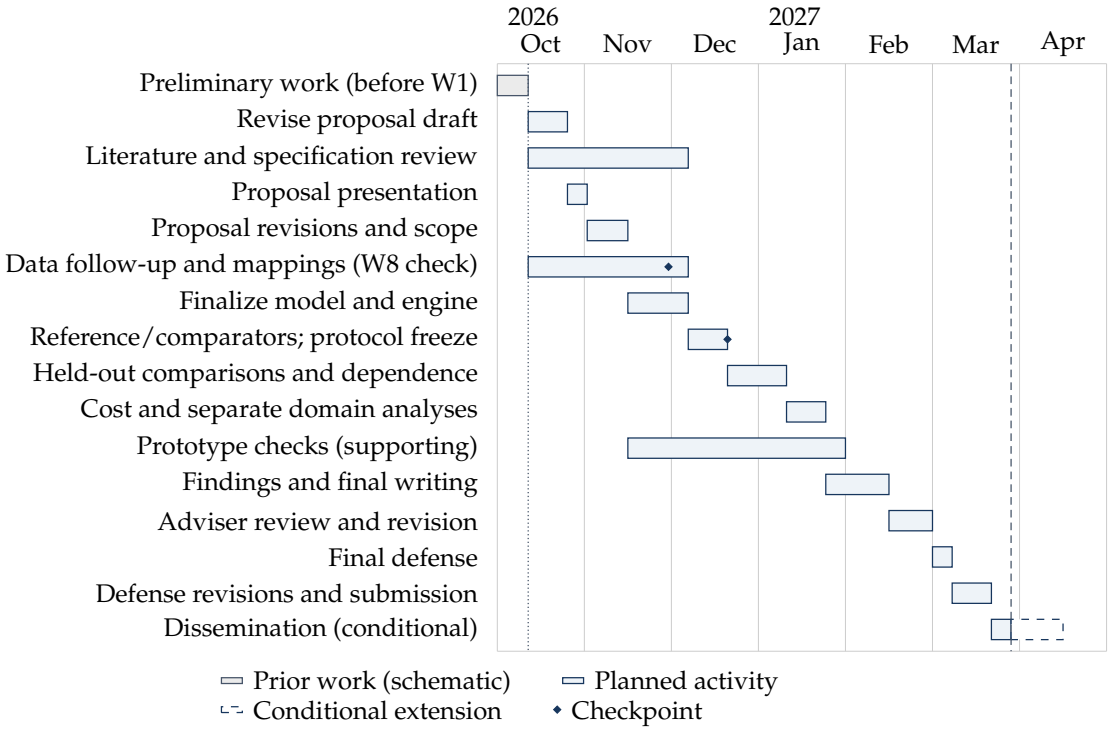



**Figure 3.4.** Provisional schedule. The dotted and dashed boundaries mark the start and end of the placeholder 24-week core (October 12, 2026–March 28, 2027). Diamonds mark the W8 data review and protocol freeze; the dashed bar shows conditional dissemination beyond the core. Presentation, defense, submission, and event dates remain subject to confirmation. The prior-work bar does not specify the original completion dates or duration.

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# APPENDIX A
# Worked Example of the Generic GBBRPM Core Architecture

## A.1. Purpose and Scope of the Example

This appendix provides a transparent, step-by-step calculation of the generic Graph-Based Bounded Risk Propagation Model (GBBRPM) core architecture. The example demonstrates source-node initialization, relation-level contribution, bounded convergence, local disturbance at a receiving node, current-state recursion, and final comparative ranking. All values are controlled, dimensionless illustrative inputs. They are not drainage measurements, software observations, electrical measurements, calibrated probabilities, or evidence of empirical predictive validity.

The calculation uses the controlled directed acyclic graph shown in Figure A.1. Its directed relations are $N_1 \to N_2$, $N_4 \to N_2$, and $N_2 \to N_3$. Nodes $N_1$ and $N_4$ are sources, the two incoming paths converge at $N_2$, and the accumulated state at $N_2$ is then propagated to $N_3$. A valid topological evaluation order is $N_1, N_4, N_2, N_3$.

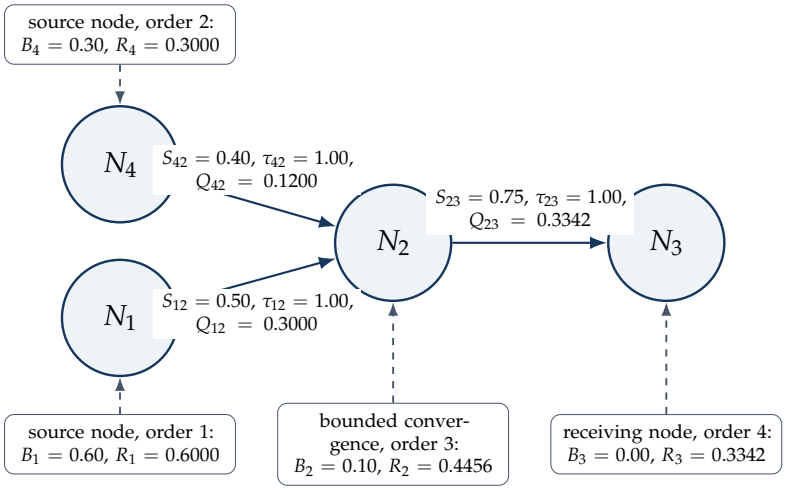



**Figure A.1.** Controlled four-node DAG used in the Appendix A calculation. Node callouts show evaluation order, local disturbance, and computed state; relation labels show susceptibility, neutral transmission, and incoming contribution.

## A.2. Controlled Inputs

Table A.1 lists the local disturbance assigned to each node. The nonzero $B_2$ value is included deliberately to show that a receiving node can contain both locally introduced and inherited risk.

**Table A.1.** Controlled Node Inputs for the Worked Example

| Node | Predecessor set $P(j)$ | Local disturbance $B_j$ |
|---|---|---|
| $N_1$ | $\emptyset$ | 0.60 |
| $N_4$ | $\emptyset$ | 0.30 |
| $N_2$ | $\{N_1, N_4\}$ | 0.10 |
| $N_3$ | $\{N_2\}$ | 0.00 |

Table A.2 lists the relation-specific susceptibilities and transmission factors. Neutral transmission $\tau_{ij} = 1$ is used consistently with the frozen v1 baseline. Therefore, the worked example does not claim to evaluate heterogeneous transmission.

**Table A.2.** Controlled Relation Inputs for the Worked Example

| Relation | Susceptibility $S_{ij}$ | Transmission $\tau_{ij}$ | Role |
|---|---|---|---|
| $N_1 \rightarrow N_2$ | 0.50 | 1.00 | First input to convergence |
| $N_4 \rightarrow N_2$ | 0.40 | 1.00 | Second input to convergence |
| $N_2 \rightarrow N_3$ | 0.75 | 1.00 | Recursive downstream transfer |

All supplied values lie in $[0, 1]$, every relation has a valid endpoint, and the graph contains no directed cycle. The inputs therefore satisfy the declared bounded-input and DAG requirements before propagation begins.

### A.3. Core Update Rules

For a directed relation $i \rightarrow j$, the incoming contribution is calculated as

$$Q_{ij} = S_{ij}\tau_{ij}R_i. \qquad \textit{(Equation A-1)}$$

where:
$Q_{ij}$ = contribution entering node $j$ from predecessor $i$;
$S_{ij}$ = bounded relation-specific susceptibility;
$\tau_{ij}$ = bounded transmission or influence factor; and
$R_i$ = current accumulated state of upstream node $i$.

The receiving-node state is calculated as

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}). \qquad \textit{(Equation A-2)}$$

where:
$R_j$ = final bounded state of receiving node $j$;
$B_j$ = local disturbance introduced directly at node $j$;
$P(j)$ = set of immediate predecessors of node $j$; and
$Q_{ij}$ = incoming contribution from predecessor $i$.

The multiplicative-complement expression is a bounded saturating aggregator. It is used algebraically and is not interpreted as a probabilistic union.

### A.4. Step 1: Initialize the Source Nodes

Nodes $N_1$ and $N_4$ have no predecessors. For either source node, the product over the empty predecessor set equals one. Equation A-2 therefore reduces to $R_j = B_j$:

$$
\begin{aligned}
R_1 &= 1 - (1 - B_1) \prod_{i \in \emptyset} (1 - Q_{i1}) \\
&= 1 - (1 - 0.60)(1) \\
&= 0.60, \\
R_4 &= 1 - (1 - B_4) \prod_{i \in \emptyset} (1 - Q_{i4}) \\
&= 1 - (1 - 0.30)(1) \\
&= 0.30.
\end{aligned}
\qquad \text{(Equation A-3)}
$$

Thus, a source node can contain a local disturbance. It does not require a special propagation rule: the ordinary aggregation rule produces its local state automatically through the empty-product convention.

### A.5. Step 2: Calculate Contributions Entering $N_2$

The current source states are now available for the two relations entering $N_2$. Applying Equation A-1 gives

$$
\begin{aligned}
Q_{12} &= S_{12} \tau_{12} R_1 \\
&= (0.50)(1.00)(0.60) \\
&= 0.30, \\
Q_{42} &= S_{42} \tau_{42} R_4 \\
&= (0.40)(1.00)(0.30) \\
&= 0.12.
\end{aligned}
\qquad \text{(Equation A-4)}
$$

The two contributions are calculated separately because they arrive through different directed relations with different susceptibilities. Their raw sum is not used as the receiving-node state; they are combined through the bounded aggregation rule in the next step.

### A.6. Step 3: Aggregate Local and Inherited Risk at $N_2$

Node $N_2$ has local disturbance $B_2 = 0.10$ and two incoming contributions. Substitution into Equation A-2 gives

$$\begin{aligned} R_2 &= 1 - (1 - B_2)(1 - Q_{12})(1 - Q_{42}) \\ &= 1 - (1 - 0.10)(1 - 0.30)(1 - 0.12) \\ &= 1 - (0.90)(0.70)(0.88) \\ &= 1 - 0.5544 \\ &= 0.4456. \end{aligned} \qquad \text{(Equation A-5)}$$

The value $R_2 = 0.4456$ represents the combined effect of the local disturbance at $N_2$ and the two inherited contributions. Although all three components contribute to the result, the multiplicative-complement form keeps the computed state within $[0, 1]$.

For comparison, removing the local disturbance while retaining both inherited contributions would give $1 - (1 - 0.30)(1 - 0.12) = 0.3840$. The increase from 0.3840 to 0.4456 is therefore caused by including the receiving node's own local disturbance through the same bounded rule rather than adding it after propagation.

### A.7. Step 4: Propagate the Current State to $N_3$

The contribution from $N_2$ to $N_3$ uses the newly accumulated state $R_2 = 0.4456$, not only the original local disturbance $B_2 = 0.10$:

$$\begin{aligned} Q_{23} &= S_{23}\tau_{23}R_2 \\ &= (0.75)(1.00)(0.4456) \\ &= 0.3342. \end{aligned} \qquad \text{(Equation A-6)}$$

Because $N_3$ has no local disturbance and only $N_2$ as its predecessor, its state is

$$\begin{aligned} R_3 &= 1 - (1 - B_3)(1 - Q_{23}) \\ &= 1 - (1 - 0.00)(1 - 0.3342) \\ &= 1 - (1.00)(0.6658) \\ &= 0.3342. \end{aligned} \qquad \text{(Equation A-7)}$$

This step demonstrates current-state recursion. If the calculation incorrectly used only $B_2$, the relation contribution would be $(0.75)(1.00)(0.10) = 0.0750$, which would discard the risk inherited by $N_2$. The implemented architecture instead propagates $R_2$, preserving the accumulated upstream effect in the downstream evaluation.

### A.8. Step 5: Produce the Comparative Ranking

After all nodes have been evaluated once in topological order, their final states are sorted in descending order. Table A.3 reports the resulting prioritization.

**Table A.3.** Final Node States and Comparative Ranking

| Rank | Node | Final state $R_i$ |
|---|---|---|
| 1 | $N_1$ | 0.6000 |
| 2 | $N_2$ | 0.4456 |
| 3 | $N_3$ | 0.3342 |
| 4 | $N_4$ | 0.3000 |

The ranking is a comparison among modeled node states under this controlled configuration. It does not mean that $N_1$ has a 60% probability of failure or that the numerical spacing between ranks has a domain-independent physical meaning. If two nodes obtain exactly equal final states, the implementation retains the tie and assigns them the same risk standing rather than applying an unsupported secondary tie-breaker.

### A.9. Calculation Trace and Interpretation

Table A.4 consolidates the complete evaluation trace. It also shows that every downstream calculation uses only predecessor states that have already been computed in the declared topological order.

**Table A.4.** Complete Topological Evaluation Trace

| Order | Node | Available predecessor states | Incoming contributions | Result |
|---|---|---|---|---|
| 1 | $N_1$ | None | Empty product | $R_1 = 0.6000$ |
| 2 | $N_4$ | None | Empty product | $R_4 = 0.3000$ |
| 3 | $N_2$ | $R_1 = 0.6000$, $R_4 = 0.3000$ | $Q_{12} = 0.3000$, $Q_{42} = 0.1200$ | $R_2 = 0.4456$ |
| 4 | $N_3$ | $R_2 = 0.4456$ | $Q_{23} = 0.3342$ | $R_3 = 0.3342$ |

The worked example establishes seven features of the generic architecture. First, the empty-product convention gives source nodes their local states without a separate initialization equation. Second, relation susceptibility modifies each upstream state before it enters the receiving node. Third, neutral transmission preserves the transmission interface without introducing unsupported relation differences. Fourth, multiple incoming contributions and a receiving-node disturbance are combined through one bounded rule. Fifth, the evaluator propagates the current accumulated state rather than only the original local input. Sixth, one topological pass is sufficient because the evaluated graph is acyclic. Seventh, prioritization is derived only after all node states have been calculated.

These calculations explain the mechanics of the frozen core architecture. They do not establish that the illustrative parameter values are appropriate for a particular domain. Drainage, software-dependency, and electrical-network instantiations require separate evidence for the meanings and construction of their local disturbances, susceptibilities, transmission factors, and outcome comparators.

# APPENDIX B
# Preliminary Worked Calculation for the Software-Dependency Case

## B.1. Purpose and Selected Scenario

This appendix traces one preliminary software-domain result through the frozen GBBRPM equations. The case uses the real Express 4.18.2 dependency topology, the frozen OSV advisory snapshot, unit susceptibility on every retained dependency relation, and the controlled uniform transmission setting $\tau = 0.50$. This setting is selected because it demonstrates both intermediate-node recursion and multi-source convergence at Express. It is a sensitivity setting, not an empirically calibrated software transmission coefficient.

The original deps.dev relation points from a dependent package to the dependency it uses. For propagation, the relation is reversed so that a local disturbance in a dependency can travel toward packages that depend on it. The calculation below therefore uses dependency-to-dependent direction. Package names are shortened for readability, but every value corresponds to the exact package version retained in the preliminary software case.

## B.2. Local Disturbance and Relation Inputs

The seven evidence-backed package versions have the local disturbances shown in Table B.1. Each relation uses $S_{ij} = 1$ and the selected uniform sensitivity value $\tau_{ij} = 0.50$.

**Table B.1.** Software Inputs Used in the Worked Calculation

| Package version | Severity | $B_i$ | Relevant outgoing relation(s) |
|---|---|---|---|
| `qs@6.11.0` | Moderate | 0.50 | `body-parser`, Express |
| `send@0.18.0` | Low | 0.25 | `serve-static`, Express |
| `body-parser@1.20.1` | High | 0.75 | Express |
| `path-to-regexp@0.1.7` | High | 0.75 | Express |
| `cookie@0.5.0` | Low | 0.25 | Express |
| `serve-static@1.15.0` | Low | 0.25 | Express |
| `express@4.18.2` | Moderate | 0.50 | Root dependent package |

Packages without advisory-backed local disturbance have $B_i = 0$. Because their upstream states are also zero in this frozen case, their contributions to the nodes calculated below are zero and their $(1 - Q_{ij})$ terms equal one. They can therefore be omitted from the displayed products without changing the result.

## B.3. Step 1: Initialize the Disturbed Dependency Nodes

The selected upstream disturbed packages have no positive incoming contribution in this scenario. Their accumulated states therefore equal their local

disturbances:

$$R_{\texttt{qs}} = 0.50, \qquad R_{\texttt{send}} = 0.25, \qquad R_{\texttt{path-to-regexp}} = 0.75, \qquad R_{\texttt{cookie}} = 0.25. \tag{\textit{Equation B-1}}$$

These are comparative normalized disturbances derived from the declared advisory-severity mapping; they are not exploitation probabilities.

### B.4. Step 2: Evaluate the Intermediate Dependent Packages

The contribution from `qs` to `body-parser` is

$$\begin{aligned} Q_{\texttt{qs,body-parser}} &= S\tau R_{\texttt{qs}} \\ &= (1.00)(0.50)(0.50) \\ &= 0.25. \end{aligned} \tag{\textit{Equation B-2}}$$

Combining this incoming contribution with the local disturbance $B_{\texttt{body-parser}} = 0.75$ gives

$$\begin{aligned} R_{\texttt{body-parser}} &= 1 - (1 - 0.75)(1 - 0.25) \\ &= 1 - (0.25)(0.75) \\ &= 0.8125. \end{aligned} \tag{\textit{Equation B-3}}$$

Similarly, the contribution from `send` to `serve-static` is

$$Q_{\texttt{send,serve-static}} = (1.00)(0.50)(0.25) = 0.125. \tag{\textit{Equation B-4}}$$

The receiving package also has local disturbance $B_{\texttt{serve-static}} = 0.25$, so

$$\begin{aligned} R_{\texttt{serve-static}} &= 1 - (1 - 0.25)(1 - 0.125) \\ &= 1 - (0.75)(0.875) \\ &= 0.34375. \end{aligned} \tag{\textit{Equation B-5}}$$

Equations B-3 and B-5 demonstrate current-state recursion: the states later transmitted to Express contain both local and inherited effects.

### B.5. Step 3: Calculate the Contributions Entering Express

At $\tau = 0.50$ and $S = 1$, the six nonzero contributions entering Express are shown in Table B.2.

**Table B.2.** Nonzero Contributions Entering Express at $\tau = 0.50$

| Dependency source $i$ | Current $R_i$ | $Q_{i,\text{Express}} = R_i(1)(0.50)$ |
|---|---|---|
| `body-parser` | 0.812500 | 0.406250 |
| `cookie` | 0.250000 | 0.125000 |
| `path-to-regexp` | 0.750000 | 0.375000 |
| `qs` | 0.500000 | 0.250000 |
| `send` | 0.250000 | 0.125000 |
| `serve-static` | 0.343750 | 0.171875 |

For example, the first contribution uses the accumulated intermediate state:

$$Q_{\texttt{body-parser},\text{Express}} = (1.00)(0.50)(0.8125) = 0.40625. \qquad \textit{(Equation B-6)}$$

Using only the original $B_{\texttt{body-parser}} = 0.75$ would instead give 0.375 and would incorrectly discard the contribution already inherited from `qs`.

### B.6. Step 4: Aggregate Local and Inherited State at Express

Express has its own advisory-derived local disturbance $B = 0.50$. Applying the multiplicative-complement rule to the six nonzero incoming contributions gives

$$\begin{aligned} R_{\text{Express}} &= 1 - (1 - 0.50) \times (1 - 0.40625)(1 - 0.125)(1 - 0.375) \\ &\qquad \times (1 - 0.25)(1 - 0.125)(1 - 0.171875) \\ &= 1 - (0.50)(0.59375)(0.875)(0.625) \\ &\qquad \times (0.75)(0.875)(0.828125) \\ &= 1 - (0.50)(0.1764643192) \\ &= 0.9117678404 \approx 0.9118. \end{aligned} \qquad \textit{(Equation B-7)}$$

This independently reproduces the preliminary result for Express at $\tau = 0.50$. At this setting Express becomes the highest-ranked node, while all states remain within $[0, 1]$. The example establishes traceability of the software instantiation and the effect of the controlled transmission sweep; it does not validate prediction of exploitation or operational incidents.

This calculation also exposes the dependence limitation. The `qs` state contributes both directly to Express and indirectly through `body-parser`; similarly, `send` contributes directly and through `serve-static`. The product in Equation (B-7) treats the corresponding inputs separately despite their shared origins. It reproduces the implemented v1 result, but does not correct or quantify overlap inflation.

# APPENDIX C
# Preliminary Worked Calculation for the Component 2 Electrical-Network Case

## C.1. Purpose and Selected Scenario

This appendix traces the preliminary Component 2 root-disturbance scenario with $B_{\text{C2_N01}} = 0.75$, measured-loading scale 1.00, and neutral transmission $\tau = 1$. It first demonstrates how phasor-derived apparent power and declared capacity produce electrical susceptibility, then follows the resulting risk through the 12-node observable arborescence. The inputs are derived model-ready summaries; restricted raw phasor records and credentials are not reproduced.

All nodes other than `C2_N01` have $B_i = 0$ in this controlled scenario. Because each non-root node in the observable Component 2 graph has one predecessor, its update reduces to $R_j = Q_{ij} = S_{ij}\tau_{ij}R_i$. This simplification is specific to the selected single-source arborescence case; it does not replace the general bounded aggregation rule.

## C.2. Step 1: Derive Baseline Electrical Susceptibility

For the source transformer `tr_117`, the 24 synchronized phasor snapshots produced mean downstream real power $P = 985.9568$ kW and reactive power $Q = 599.3408$ kvar. The loading is the mean of the apparent powers computed at each snapshot, rather than the apparent power computed from these two means. Writing the snapshot powers as $P_t$ and $Q_t$,

$$\begin{aligned} L_{12} &= \frac{1}{24}\sum_{t=1}^{24}\sqrt{P_t^2 + Q_t^2} \\ &= 1153.8297 \text{ kVA}. \end{aligned} \qquad \text{(Equation C-1)}$$

For comparison, $\sqrt{\bar{P}^2 + \bar{Q}^2} \approx 1153.8285$ kVA. The small difference arises from the order of averaging and the nonlinear square-root operation; the snapshot-mean loading used below is unchanged.
Using the declared transformer capacity $C_{12} = 3750$ kVA, the source-edge susceptibility is

$$\begin{aligned} S_{12} &= \min\left(1, \frac{L_{12}}{C_{12}}\right) \\ &= \min\left(1, \frac{1153.8297}{3750}\right) \\ &= 0.3076879. \end{aligned} \qquad \text{(Equation C-2)}$$

For comparison, `line_408` has $L = 705.9977$ kVA and $C = 1787.94$ kVA, giving

$$S_{23} = \min\left(1, \frac{705.9977}{1787.94}\right) = 0.3948666. \qquad \text{(Equation C-3)}$$

The same rule is applied once to every observable edge. Table C.1 lists the model-ready values needed for the complete trace.

**Table C.1.** Component 2 Edge Inputs Used in the Worked Calculation

| Relation | Element | $L$ (kVA) | $C$ (kVA) | $S = L/C$ |
|---|---|---|---|---|
| C2_N01 → C2_N02 | `tr_117` | 1153.8297 | 3750.00 | 0.3076879 |
| C2_N02 → C2_N03 | `line_408` | 705.9977 | 1787.94 | 0.3948666 |
| C2_N02 → C2_N11 | `line_409` | 375.4093 | 1787.94 | 0.2099675 |
| C2_N03 → C2_N04 | `line_415` | 235.8730 | 1787.94 | 0.1319244 |
| C2_N03 → C2_N06 | `tr_121` | 317.7301 | 1000.00 | 0.3177301 |
| C2_N03 → C2_N07 | `line_417` | 153.8553 | 1787.94 | 0.0860517 |
| C2_N03 → C2_N09 | `line_416` | 24.9112 | 1787.94 | 0.0139329 |
| C2_N04 → C2_N05 | `tr_119` | 235.8730 | 1500.00 | 0.1572487 |
| C2_N07 → C2_N08 | `tr_123` | 153.8553 | 750.00 | 0.2051404 |
| C2_N09 → C2_N10 | `tr_125` | 24.9112 | 1000.00 | 0.0249112 |
| C2_N11 → C2_N12 | `tr_127` | 375.4093 | 1500.00 | 0.2502728 |

All listed ratios are below one, so clipping does not alter them in the baseline scenario. The values are operating loading-to-capacity ratios, not failure probabilities.

### C.3. Step 2: Initialize the Root and Evaluate the First Transfer

The root has no predecessor, so the empty-product convention gives

$$R_{\text{C2_N01}} = B_{\text{C2_N01}} = 0.75. \qquad \text{(Equation C-4)}$$

The contribution through `tr_117` is

$$\begin{aligned} Q_{12} &= S_{12}\tau_{12}R_1 \\ &= (0.3076879)(1.00)(0.75) \\ &= 0.2307659. \end{aligned} \qquad \text{(Equation C-5)}$$

Because $B_{\text{C2_N02}} = 0$ and the node has only this predecessor,

$$R_{\text{C2_N02}} = 1 - (1 - 0)(1 - 0.2307659) = 0.2307659. \qquad \text{(Equation C-6)}$$

### C.4. Step 3: Propagate Through the Main Branch

Using the newly accumulated state at `C2_N02`, the contribution to `C2_N03` is

$$
\begin{aligned}
Q_{23} &= (0.3948666)(1.00)(0.2307659) \\
&= 0.0911218, \\
R_{\text{C2_N03}} &= 0.0911218.
\end{aligned}
\qquad \text{(Equation C-7)}
$$

The four outgoing branches from `C2_N03` are then evaluated independently. For example, the transformer branch to `C2_N06` gives

$$
\begin{aligned}
Q_{36} &= (0.3177301)(1.00)(0.0911218) \\
&= 0.0289521, \\
R_{\text{C2_N06}} &= 0.0289521.
\end{aligned}
\qquad \text{(Equation C-8)}
$$

The parallel branch from `C2_N02` to `C2_N11` gives

$$
\begin{aligned}
R_{\text{C2_N11}} &= (0.2099675)(1.00)(0.2307659) \\
&= 0.0484533, \\
R_{\text{C2_N12}} &= (0.2502728)(1.00)(0.0484533) \\
&= 0.0121266.
\end{aligned}
\qquad \text{(Equation C-9)}
$$

These calculations show why risk decreases differently across branches: every edge applies its own measured-loading susceptibility to the current upstream state.

### C.5. Step 4: Complete the Topological Trace

Table C.2 reports the complete one-pass trace. Minor differences in the displayed products can result from rounding; the evaluator uses the unrounded input values listed in the model tables.

**Table C.2.** Component 2 Root-Severity Trace at $B_{N01} = 0.75$

| Depth | Node | Calculation using unrounded predecessor state | Final $R_i$ |
|---|---|---|---|
| 0 | C2_N01 | Local disturbance | 0.7500000 |
| 1 | C2_N02 | $0.75(0.3076879)$ | 0.2307659 |
| 2 | C2_N03 | $R_{N02}(0.3948666)$ | 0.0911218 |
| 2 | C2_N11 | $R_{N02}(0.2099675)$ | 0.0484533 |
| 3 | C2_N04 | $R_{N03}(0.1319244)$ | 0.0120212 |
| 3 | C2_N06 | $R_{N03}(0.3177301)$ | 0.0289521 |
| 3 | C2_N07 | $R_{N03}(0.0860517)$ | 0.0078412 |
| 3 | C2_N09 | $R_{N03}(0.0139329)$ | 0.0012696 |
| 3 | C2_N12 | $R_{N11}(0.2502728)$ | 0.0121266 |
| 4 | C2_N05 | $R_{N04}(0.1572487)$ | 0.0018903 |
| 4 | C2_N08 | $R_{N07}(0.2051404)$ | 0.0016085 |
| 4 | C2_N10 | $R_{N09}(0.0249112)$ | 0.0000316 |

Summing the unrounded states gives

$$\sum_{i=1}^{12} R_i = 1.1860821582 \approx 1.186, \qquad \text{(Equation C-10)}$$

which reproduces the risk sum retained in the preliminary electrical case. The three nodes above the declared 0.05 affected-node threshold are `C2_N01`, `C2_N02`, and `C2_N03`; the maximum terminal-node state is $R_{\text{C2_N06}} = 0.0289521 \approx 0.029$.

The calculation establishes that measured normal-operation loading can be converted into heterogeneous bounded susceptibilities and evaluated by the unchanged GBBRPM core. The injected $B = 0.75$ is controlled rather than an observed electrical fault, so the result is mechanism and sensitivity evidence rather than event-outcome validation.

# APPENDIX D
# Selected Preliminary Computational Evidence

## D.1. Status, Purpose, and Provenance

This appendix preserves selected completed exploratory findings that motivated the proposed evaluation. These outputs were reported in the pre-consultation full-study manuscript and have not been regenerated under every newly specified proposal convention. They demonstrate implementation feasibility and expose limitations; they are not the final results of the proposed study.

The preliminary core used reconstructed N1–N5 fixture inputs and a frozen formulation. The restored scenario inventory contained 136 runs. The five controlled fixtures contained 6, 8, 8, 7, and 20 nodes, respectively. The software topology and measured electrical loading are real-data inputs, while their disturbance mappings or scenario injections remain distinct from independent incident outcomes. A complete pre-consultation PDF and the original chapter sources are retained in the accompanying research archive.

## D.2. Controlled Behavior and Comparator Differences

The baseline outlet states were 0.053760, 0.408000, 0.106814, 0.351372, and 0.708640 for N1–N5. These values are comparative indices under the declared fixtures, not failure probabilities.

**Table D.1.** Preliminary N5 comparator agreement with heterogeneous GBBRPM.

| Comparator | Spearman $\rho$ | Top-three Jaccard |
|---|---|---|
| Local-only | 0.108627 | 0.0 |
| Uniform $S = 0.25$ | -0.042105 | 0.0 |
| Uniform $S = 0.50$ | 0.150376 | 0.0 |
| Uniform $S = 0.75$ | 0.696241 | 0.5 |

These differences show that the supplied relation conditions and propagated context changed the ordering. Agreement with GBBRPM is not a ground-truth accuracy measure, and the table does not establish that GBBRPM is operationally superior to the comparison scores.

## D.3. Preliminary Property, Robustness, and Dependence Checks

The prior report recorded 600 randomized property-test trials with no reported failure in its boundedness, local-disturbance monotonicity, $B = 1$ boundary, and zero-susceptibility checks. This is finite implementation evidence alongside the analytical arguments, not a proof of all possible extensions or implementations.

**Table D.2.** Preliminary N5 robustness summary, 600 trials per perturbation level.

| Perturbation | Mean Spearman | 5th percentile | 95th percentile | Mean top-three Jaccard | Mean top-20% Jaccard |
|---|---|---|---|---|---|
| ±5% | 0.979190 | 0.962331 | 0.992481 | 0.744500 | 0.776000 |
| ±10% | 0.938267 | 0.886289 | 0.974436 | 0.546667 | 0.604873 |
| ±20% | 0.845552 | 0.697067 | 0.935338 | 0.388833 | 0.440823 |

These previously reported set conventions are kept distinguishable from the explicit cutoff-tie protocol for planned runs. The top-20% metric has nominal $k = 4$ for N5. Small perturbations produced broadly stable ordering, while the highest-priority membership was more sensitive.

The ancestry-aware path-pruned diagnostic gave outlet-state differences of 0.000000 for N3, 0.099372 for N4, and 0.322637 for N5 relative to ordinary aggregation. This is sensitivity to a declared counterfactual, not a validated estimate of true physical bias. It demonstrates why a bounded output and stable ranking do not establish dependence correction.

### D.4. Preliminary Runtime Evidence

The existing sparse-DAG experiment reported median timings from 1.628378 ms for 20 nodes and 60 edges to 16.650025 ms for 1000 nodes and 3000 edges, after five warm-ups and 30 repetitions. These implementation-specific timings do not provide a matched speed comparison with a hydraulic simulator, memory evidence, or a breakdown of validation, propagation, and sorting. The planned cost protocol will supply those distinctions rather than infer them from this table range.

### D.5. Preliminary Hydraulic-Reference Disagreement

The prior 12-scenario SWMM comparison reported the following:

**Table D.3.** Preliminary comparison with simulated hydraulic worsening.

| Hydraulic outcome | Mean Spearman | Top-20% Jaccard | Positive worsening outside downstream scope |
|---|---|---|---|
| Maximum depth | -0.416 | 0.119 | 71.2% |
| Flooding volume | -0.485 | 0.000 | 40.5% |

The negative associations and low priority-set overlaps do not support hydraulic-outcome ranking equivalence under this protocol. Outside-scope worsening is consistent with mechanisms omitted by the downstream-only DAG index, but the comparison does not isolate a causal hydraulic mechanism in each

scenario. Neither disagreement nor low computation cost validates drainage inspection or maintenance utility.

### D.6. Preliminary Software and Electrical Instantiation

The Express 4.18.2 case used 71 packages, 128 directed relations, and seven advisory-backed local disturbances. Root risk increased from 0.5000 at $\tau = 0$ to 0.9975 at $\tau = 1$, and its rank became first by $\tau = 0.50$. Appendix B traces the intermediate setting. Shared `qs` and `send` paths may inflate the root state; their contribution is not quantified by the stable ranking.

The observable Component 2 case used 12 nodes, 11 edges, and loading derived from 24 aligned phasor snapshots. The 43 controlled scenarios tested severity, location, load scaling, transmission, and multiple local sources. The root-disturbance case $B_{\mathrm{C2_N01}} = 0.75$ gave risk sum approximately 1.186. Appendix C provides its trace. These are injected-disturbance experiments on normal-operation measurements, not independent event-response prediction.

### D.7. Preliminary Prototype Feasibility

The prior implementation report recorded a controlled-fixture inventory, frozen N5 reproduction, valid imported-graph evaluation, invalid-input rejection, and a production frontend build. Those checks support implementation feasibility within the tested configuration. The planned study will rerun and extend engine/prototype consistency checks; it will not infer usability, security, field applicability, or operational readiness from the preliminary functional tests.

The member-supplied prototype update reports a stored trace contract, optional concurrent replay, and checks of record coverage, schedule dependencies, input non-mutation, and settled readout agreement. The proposed verification in Chapter 3 will address these items using a recorded prototype revision and retained test evidence.

# APPENDIX E
# Supporting Prototype Contract and Planned Verification

## E.1. Scope and Trace Contract

This is a supporting design and verification plan, not a completed test report. Prototype work will be limited to input validation, engine integration, and inspection of evaluated quantities. The planned mechanism experiments will take priority over additional presentation features.

Synthetic and imported DAGs will share one schema. Explicit susceptibility will be displayed as supplied. The service will return its topological order and engine-computed $B$, $Q$, and $R$ quantities, with one source-or-aggregate settlement record per node and one transfer record per edge. A receiving node may appear in several transfer records. Trace assembly will not repeat the propagation evaluation, but its processing and storage costs will be recorded.

## E.2. Presentation Boundary

Optional replay will present stored results without changing inputs, final states, or rankings. Independent branches may progress concurrently; a receiving state will settle after its incoming contributions arrive. Interpolated readouts will be presentation values, checked against engine quantities only after settlement. Display timing will not represent physical propagation speed. Supported completion, cancellation, and reduced-motion behavior will be documented for the tested revision.

## E.3. Verification and Outstanding Evidence

The verification record will retain engine/prototype revisions, fixtures, expected quantities, tolerances, and test evidence. It will cover N1–N5 numerical agreement and trace coverage, selected receiving-state reconstruction, valid imports, invalid-input rejection, predecessor-settlement order, concurrent branches, input non-mutation, and settled-value agreement across display modes. Frontend build and edit/run checks will be complemented by automated and manual browser inspection recorded for the tested revision. Functional checks will not establish operational utility or deployment readiness.